COMSATS University Islamabad,

Sahiwal Campus, Pakistan.

AI Mock Interview Platform

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Bachelor of Science in Computer Science (2022-2026)

COMSATS University Islamabad,
Sahiwal Campus, Pakistan.

AI Mock Interview Platform

A Project Presented To

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In partial fulfillment

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Bachelor of Science in Computer Science (2022-2026)

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Asma Murtaza	**Areeba Tariq**	**Eman Arshad**
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Certificate Of Approval

It is to certify that the final year project of BS(CS) **“AI Mock Interview Platform”** was developed by **Asma Murtaza (CIIT/FA22-BCS-133/SWL)**,**Areeba Tariq (CIIT/FA22-BCS-139/SWL)**, **Eman Arshad (CIIT/FA22-BCS-194/SWL)** under the supervision of “ **Dr. Yawar Abbas Abid**” and that in their opinion, it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Science.

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Executive Summary

AI Mock Interview Platform (AMIP) is an innovative AI-powered system designed to help job seekers, students, and professionals prepare effectively for interviews and boost their confidence in the hiring process. Unlike traditional preparation methods, it offers a dynamic and interactive environment where users can practice unlimited mock interviews, receive personalized feedback, and continuously improve their performance.

The platform emphasizes personalization by allowing candidates to customize interviews based on job role, experience level, and technologies. Through its AI-driven voice agent, the system delivers realistic conversations that replicate the pressure and flow of actual interviews. This ensures that users not only rehearse answers but also build confidence in handling diverse and challenging questions.

Unlike traditional preparation methods, the platform integrates a feedback module that goes beyond generic responses. It highlights weak areas, evaluates cultural fit, and provides actionable insights for improvement. This structured feedback loop enables candidates to enhance their performance and track progress across multiple sessions.

Technically, the AI Mock Interview Platform is built using modern frameworks such as Next.js, Tailwind CSS, Firebase, and VAPI, ensuring secure authentication, responsive design, and scalability. The platform supports real-time performance, lifelike AI conversations, and a seamless user experience across devices, making interview preparation accessible anytime and anywhere.

The project follows an Agile development model, incorporating iterative improvements based on user feedback. Comprehensive testing, including unit, integration, and user acceptance testing, ensures reliability and accuracy across all features. By combining modern web technologies with AI-driven interview preparation, the platform empowers candidates to overcome anxiety, refine their skills, and confidently pursue their career goals.

Acknowledgment

All praise is to almighty Allah who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor **"Dr. Yawar Abbas Abid".** Without their supervision, advice, and valuable guidance, the completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty and hard work.

Abbreviations

Abbreviation	Full Form
2FA	Two-Factor Authentication
AMIP	AI Mock Interview Platform
AI	Artificial Intelligence
API	Application Programming Interface
BSc	Bachelor of Science
CI/CD	Continuous Integration / Continuous Deployment
COMSATS	COMSATS University Islamabad
Firebase	Google Firebase (Backend-as-a-Service platform)
Gemini	Google Gemini
FT	Functional Testing
IT	Integration Testing
Next.js	Next.js (React framework for server-rendered apps)
React	React (JavaScript library for UI)
Shadcn	Shadcn UI (Reusable React component library using Tailwind CSS)
Tailwind	Tailwind CSS (utility-first CSS framework)
TS	TypeScript
UC	Use Case
UI/UX	User Interface / User Experience
UT	Unit Testing
VAPI	Voice API (used for AI voice agent integration)
Vercel	Vercel (deployment platform)
Zod	Zod (TypeScript Validation Library)

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Chapter 1
Introduction

1. Introduction

The AI Mock Interview Platform is a modern, web-based application designed to empower final-year students and fresh graduates as they prepare to enter the professional world. In today's competitive job market, many candidates face challenges such as interview anxiety, lack of practical exposure, and difficulty in presenting their skills effectively. PrepWise addresses these issues by providing a unified space for interview simulation, personalized feedback, and skill refinement.

Unlike traditional preparation methods or scattered online resources, the AI Interview Platform enables users to engage in lifelike AI-driven interview sessions. Through its integrated Voice AI module (VAPI), candidates experience realistic conversations that replicate the flow of professional interviews. The system evaluates communication clarity, confidence, and subject knowledge, offering structured feedback to help users continuously improve their performance.

Developed using Next.js, Tailwind CSS, TypeScript, and Firebase [1].the AI Interview Platform ensures a secure, scalable, and user-friendly environment accessible across devices. The platform follows the Agile development methodology, incorporating iterative improvements based on user feedback. While AI is used to simulate interviews and generate predictive responses, the AI Interview Platform avoids biased scoring systems, ensuring fairness and transparency for all candidates.

For recruiters, the AI Interview Platform offers a potential extension in the form of a dedicated portal to explore candidate readiness and connect directly with talent. This human-centered approach preserves recruiter judgment while fostering meaningful engagement between students and employers.

Security and reliability are ensured through authentication mechanisms (including 2FA), structured data handling, and comprehensive testing at every stage, including unit testing, functional testing, and integration testing. Developed as part of the Bachelor of Science in Computer Science (2022–2026) program at COMSATS University Islamabad, Sahiwal Campus, the AI Interview Platform is designed to be deployment-ready and capable of evolving through real-world usage. The platform aims to build confidence, enhance interview readiness, and create a direct bridge between academic achievement and early-career opportunities.

1.1 Brief Overview

The AI Mock Interview Platform aims to develop a web-based solution designed exclusively for final-year students and fresh graduates to enhance how they prepare for interviews, refine communication skills, and build confidence before entering the professional world. The system is developed using Next.js, Tailwind CSS, TypeScript, Firebase, and VAPI [2] for the front end and backend integration, ensuring a secure, scalable, and high-performance experience. The project adopts the Agile software development methodology, enabling iterative and structured development [5].

The AI Interview Platform is a unique and focused solution designed to provide users with an intelligent space where they can engage in AI-driven mock interviews, receive personalized feedback, and continuously improve their readiness for real-world opportunities. Unlike traditional preparation methods, the AI Interview Platform stands out due to its specialized focus on interview simulation and skill refinement, making it a dedicated hub for learners and job seekers. It addresses challenges such as interview anxiety, scattered preparation resources, and lack of practical exposure by enabling users to practice in lifelike scenarios, leverage Voice AI predictions, and access structured evaluations that ensure they remain confident and prepared for career opportunities aligned with their academic journey .

1.2 Relevance to Course Modules

The development of PrepWise – AI Mock Interview Platform directly applies and integrates various concepts and skills acquired throughout our Computer Science course.

i. Technical Skills:

The project involves the application of multiple programming languages, frameworks, and tools learned during the course, including Next.js, Tailwind CSS, TypeScript, and Firebase. These skills were essential for building the mock interview system and implementing core functionalities such as user authentication, interview simulation, and feedback generation. The principles of software engineering such as modular design, component-based architecture, and secure coding practices were applied to ensure system reliability, scalability, and maintainability.

ii. Software Engineering Principles:

The project adheres to software engineering principles covering requirement analysis, system design, software architecture, testing, and agile development. Each phase of PrepWise development follows structured practices to ensure scalability, maintainability, usability, and security. The system integrates concepts from web development, distributed systems, version control, and cloud deployment, making it a comprehensive computer science application.

iii. Database Design:

A strong understanding of database management was essential in designing the Firebase database for storing user profiles, interview sessions, and feedback records. The platform makes use of structured collections, relational mapping, indexing, and secure data handling to ensure fast and efficient retrieval of candidate information across large datasets.

iv. System Architecture:

The project required an understanding of modern web system architecture and cloud-based deployments. Concepts such as component-based design in Next.js, REST API integration, and microservices-ready structure were applied to build a structured and scalable system. Deployment tools such as CI/CD pipelines and cloud hosting ensure that the platform can scale efficiently and maintain high availability.

v. UI/UX Design:

A user-friendly interface is essential for the success of an interview preparation platform. Concepts from human-computer interaction and UI/UX design courses guided the creation of clean, responsive layouts using Tailwind CSS. These principles helped build an intuitive and visually consistent interface that students and recruiters can use comfortably across devices.

vi. Project Management:

Project management concepts played a critical role in planning, executing, and controlling the development phases of the AI Interview Paltform. Agile methodology supported iterative development, sprint-based progress, and continuous improvement. This ensured that features such as mock interview modules, recruiter dashboards, and feedback systems were delivered in a timely and structured manner.

vii. Networking and AI Integration:

The AI Interview Platform includes AI-driven features such as lifelike interview simulations, predictive responses, and structured feedback, which rely on concepts from machine learning and Voice AI integration (VAPI). Additionally, real-time updates such as interview session handling and feedback delivery depend on efficient API communication and backend networking principles. Understanding HTTP protocols, REST standards, and asynchronous operations ensures seamless interaction across the platform.

1.3 Project Background

The AI Mock Interview Platform was conceptualized in October 2025 at COMSATS University Islamabad, Sahiwal Campus, under the supervision of Dr. Yawar Abbas as part of the Bachelor of Science in Computer Science (2022–2026) final year project. The idea originated from the real difficulties faced by fresh graduates and final-year students during their transition into the professional world. Many students shared that they felt unprepared for job interviews, especially when experiencing them for the first time, and struggled to understand industry expectations. Recruiters similarly expressed that evaluating fresh graduates was often challenging due to their lack of interview practice, limited exposure to professional communication, and absence of a structured preparation platform.

A survey conducted with a group of students, faculty members, and recruiters revealed widespread issues in early-career preparation. Students reported a lack of confidence in interviews, difficulty explaining their academic projects, and confusion about how to present themselves professionally. Many highlighted that traditional preparation methods such as reading guides or watching tutorials failed to replicate the pressure and flow of real interviews. Recruiters, on the other hand, noted that fresh graduates often struggled with communication clarity, confidence, and practical readiness, making it difficult to assess their true potential.

The AI Interview Platform platform was designed to address these concerns by providing a dedicated space where students can practice interviews through an AI-assisted mock interview module and receive constructive feedback. The system helps students experience realistic interview scenarios, improve communication clarity, and build confidence. Unlike automated ranking or biased scoring systems, The AI Interview Platform ensures fairness and transparency by focusing on personalized feedback rather than algorithmic judgments. Recruiters benefit from the potential extension of a streamlined interface where they can explore candidate readiness and connect with potential talent based on their performance and skills.

Developed using Next.js, Tailwind CSS, TypeScript, Firebase, and VAPI, and following the Agile methodology,[6] the project emphasizes secure authentication, efficient data management, and an intuitive user experience suitable for individuals from all technical backgrounds. Over a structured development timeline, the platform progressed through requirement analysis, design, implementation, testing, and deployment phases, ensuring reliability and usability. The AI Interview Platform aligns with the university's mission to encourage innovation and practical skill development, offering a scalable solution that enhances student visibility, strengthens interview readiness, and facilitates meaningful engagement between graduates and recruiters.

1.4 Literature Review

Table 1.1 states that the AI Mock Interview Platform builds upon foundational and contemporary research in artificial intelligence, voice-based interaction systems, communication training, and interview preparation tools. This review synthesizes key studies and frameworks that validate the need for an AI-integrated, intelligent mock interview system as given in Table 1.1.

Table 1.1 Literature Review

Application Name	Weakness	Proposed Project Solution (PrepWise – AI Mock Interview Platform)
Traditional Interview Preparation Methods	Static guides, lack of real-time practice, no personalized feedback.	AI-driven mock interviews with lifelike simulations, structured feedback, and confidence-building exercises.
Professional Networking Platforms (e.g., LinkedIn)	Focused on career networking, not interview skill development; limited tools for communication training.	Dedicated environment for students and graduates to practice interviews, improve communication clarity, and receive targeted feedback.
Online Tutorials and Courses	One-way learning, no interactive practice, limited assessment of confidence or clarity.	Interactive voice-based interview sessions using VAPI, with real-time evaluation of communication and subject knowledge.
University Career Services	Limited reach, manual scheduling, lack of scalable tools for continuous practice.	Scalable web-based platform accessible anytime, offering repeated practice sessions and recruiter-focused extensions.

Job Portals / Recruitment Platforms	Generic candidate ranking, no interview readiness evaluation, no communication analysis.	Potential recruiter dashboard to explore candidate readiness, connect with talent, and assess interview performance beyond static resumes.

1.5 Analysis from Literature Review

This section evaluates findings from the literature related to interview preparation, early-career recruitment challenges, communication training, and UI/UX best practices. The reviewed research emphasizes the importance of structured interview practice, student preparedness for professional communication, and platforms designed specifically for fresh graduates. Studies highlight that students benefit most from systems that offer clarity, guided practice, and opportunities to build confidence before facing real recruiters. Research also underscores the limitations of traditional methods such as static guides, online tutorials, and scattered career services, suggesting the need for centralized platforms that support interview readiness without overwhelming technical complexity.

Unlike platforms relying on automated ranking or algorithmic scoring, the literature supports solutions that enhance user confidence, improve communication skills, and promote transparent recruiter–candidate engagement. These insights justify focus on AI-assisted mock interview practice, structured feedback, and recruiter-focused extensions, ensuring students gain both confidence and visibility as they transition into professional environments.

1.5.1 Interview Preparation Platforms

Existing interview preparation platforms generally offer static guides, recorded tutorials, or limited practice modules. Many lack the ability to comprehensively prepare a student for real interview scenarios, as they do not provide lifelike simulations, structured feedback, or confidence-building exercises. These platforms often fail to address communication clarity, stress management, and recruiter–candidate interaction in an organized and professional manner. PrepWise addresses this gap by enabling users to practice detailed, AI-assisted mock interviews tailored to their field of study and early-career goals. The platform emphasizes clarity, structured practice, and user-friendly interaction, allowing students to strengthen their communication skills, build confidence, and presents themselves more effectively in professional interviews. Unlike traditional methods, The AI Interview Platform integrates voice-based simulations and real-time feedback, ensuring that students gain practical readiness and recruiters can explore candidate performance beyond static resumes or generic rankings.

1.5.2 Professional Networking Platforms

Professional networking platforms such as LinkedIn serve broad professional audiences, mixing corporate achievements with academic experiences, but they are not specifically designed for students entering the job market for the first time and lack structured tools for interview practice, communication training, and confidence-building. The AI Interview Platform addresses this gap by offering a dedicated environment focused on interview readiness, enabling students and fresh graduates to practice lifelike AI-assisted mock interviews, improve communication clarity, and receive constructive feedback tailored to their academic background and career goals, ensuring that they build confidence and professional preparedness without distraction from unrelated networking content.

1.5.3 Online Tutorials and Interview Courses

Traditional online tutorials and interview courses provide static, one-way learning experiences that lack interactivity and do not help students practice real interview scenarios. They fail to offer personalized guidance, structured feedback, or tools to build confidence, leaving fresh graduates unsure about how to communicate their skills effectively in professional settings. The AI Interview Platform enhances this space by offering interactive, AI-assisted mock interview sessions using voice-based simulations (VAPI), enabling students to practice responses, strengthen communication clarity, and receive constructive feedback in real time. This approach helps students build confidence and readiness for interviews, ensuring they can present themselves professionally and effectively.

1.5.4 University Career Services or LMS Systems

Although LMS platforms and university career services provide access to course materials and limited career guidance, they do not include tools for structured interview practice, communication clarity, or confidence-building. These systems are primarily academic in nature and lack scalable solutions for preparing students for real-world recruitment challenges. The AI Interview Platform fills this gap by offering an AI-driven mock interview environment where students can repeatedly practice lifelike interview sessions, receive structured feedback, and strengthen their communication skills. The platform emphasizes accessibility, clarity, and professional readiness, ensuring that fresh graduates can prepare effectively for interviews beyond the limitations of traditional university services.

1.5.5 Job Portals and Recruitment Platforms

Traditional job portals offer generic candidate matching and ranking systems that prioritize prior work experience over interview readiness or communication skills. They often fail to evaluate a student's confidence, clarity, and ability to perform in real interview scenarios, leaving fresh graduates at a disadvantage. The AI Interview Platform addresses this gap by introducing the concept of a recruiter-focused dashboard that goes beyond static resumes. Through AI-assisted mock interview performance tracking, communication analysis, and structured feedback, recruiters can gain deeper insights into a candidate's readiness. This approach makes emerging talent more visible and accessible, ensuring that students are evaluated not only on past experience but also on their preparedness for professional interviews.

1.6 Methodology and Software Lifecycle

The AI Mock Interview Platform is a software project with evolving requirements, making flexibility and collaboration essential. Agile methodology is the best fit as it supports iterative growth and continuous refinement. It allows the system to adapt quickly to students, graduates, and recruiters needs. Through Agile, the AI Interview Platform simulations, feedback mechanisms, and recruiter-focused features are effectively.

1.6.1 Rationale Behind the Selected Agile Model

Agile emphasizes adaptability, teamwork, and ongoing enhancement, making it well-suited for a platform like The AI Interview Platform that requires frequent updates and user-driven improvements. Since the system involves continuous interaction with users, such as refining mock interview simulations, updating feedback mechanisms, and improving recruiter-focused features, the Agile model ensures the platform evolves in sync with real-world academic and professional needs.

Agile is based on the following principles:

1. **Iterative development:** Core features such as AI-driven mock interviews, feedback modules, and recruiter dashboards are developed in cycles, with each iteration refining functionality.
2. **Incremental delivery:** Modules are released in small, usable increments, enabling early access to features like interview simulations, performance tracking, and recruiter insights.
3. **Continuous feedback:** Students and recruiters provide ongoing input regarding usability, communication clarity, and interview relevance.

4. **Adaptive planning:** The development team refines plans based on feedback, evolving recruitment workflows, and AI model behavior, ensuring PrepWise remains aligned with user expectations.

1.7 Summary

The AI Mock Interview Platform, initiated in October 2025 at COMSATS University Islamabad, Sahiwal Campus, was developed to address the practical challenges faced by fresh graduates, final-year students, and recruiters in preparing for and evaluating interview readiness. Traditional methods such as scattered online tutorials, generic job portals, and limited university career services often fail to provide structured, interactive, and confidence-building preparation. Existing platforms like LinkedIn and online courses either lack academic depth, mix corporate and academic content, or do not provide lifelike interview practice.

To overcome these limitations, the AI Interview Platform provides a centralized, user-friendly system where students can engage in AI-assisted mock interviews, receive constructive feedback, and strengthen their communication and confidence before appearing in real interviews. For recruiters, the platform offers a dedicated interface that allows them to explore candidate performance, assess communication clarity, and connect directly with potential graduates without relying solely on static profiles or algorithmic scoring.

Developed using the Agile methodology, the AI Interview Platform emphasizes iterative development, continuous feedback, and adaptive planning to ensure the platform meets real user needs. Through modern web technologies, secure authentication, and an accessible interface, the system supports smooth interview practice, reliable data handling, and an environment that enhances student visibility and improves early-career readiness. Ultimately, PrepWise serves as a comprehensive digital space for interview preparation, confidence-building, and meaningful recruiter–candidate interaction.

Chapter 2
Problem Definition

2 Problem Definition

This chapter discusses the precise problem to be solved and the intended outcome.

2.1 Problem Statement

Students and fresh graduates often struggle to prepare effectively for interviews due to the absence of a centralized and dedicated platform. Traditional methods such as static tutorials, scattered online resources, generic job portals, and limited university career services fail to provide an academic focused environment for structured interview practice. As a result, students face difficulties in building confidence, improving communication clarity, and presenting their skills in a professional manner, leading to incomplete readiness when pursuing internships, research roles, or entry-level job opportunities. Recruiters also find it challenging to assess fresh candidates when interview preparedness and communication abilities are not clearly demonstrated.

To address these challenges, the AI Mock Interview Platform is being developed as a unified interview preparation and early-career readiness system. The AI Interview Platform enables students to engage in lifelike, AI-assisted mock interviews, receive constructive feedback, and strengthen their confidence before appearing in real hiring processes. For recruiters and institutions, the AI Interview Platform provides a dedicated dashboard where they can explore candidate performance, assess communication clarity, and directly connect with potential hires based on interview readiness rather than static profiles.

2.2 Deliverables and Development Requirements

The AI Mock Interview Platform project's deliverables and development requirements encompass a range of components essential for the successful implementation of an intelligent interview preparation and recruiter support system. These components are detailed below:

2.2.1 User Interface (UI) Design

Develop a clean, intuitive, and interview-focused interface that allows students to easily attempt mock interviews, view feedback, and track progress. The interface should be visually appealing, responsive across devices, and accessible, ensuring smooth navigation and user engagement.

2.2.2 User Registration and Profiles

Implement a secure registration and login system where students can create customizable profiles. Student profiles will include education, skills, and interview performance history, while recruiters will have role-based access to candidate dashboards.

2.2.3 Mock Interview Module

Provide a structured AI-driven mock interview system where students can practice lifelike scenarios, receive instant feedback, and track improvements. The module should support multiple interview formats, real-time evaluation, and performance analytics.

2.2.4 AI-Powered Insights and Recommendations

Integrate NLP and machine learning models to analyze student responses, communication clarity, and confidence levels. The AI engine should generate personalized recommendations, highlight strengths and weaknesses, and refine insights over time.

2.2.5 Recruiter Dashboard

Develop an advanced recruiter module featuring candidate performance analytics, skill-gap identification, and fine-grained filtering. Recruiters should be able to shortlist, bookmark, and connect with candidates based on interview readiness.

2.2.6 Security and Privacy

Ensure strong data protection using secure authentication, encrypted data handling, and role-based access control. Compliance with modern security standards and regular audits will safeguard sensitive student and recruiter information.

2.2.7 Scalability

Architect the platform for scalability using modern frameworks and containerization tools, ensuring support for increasing users, interview sessions, and recruiter activity without performance degradation.

2.2.8 Web Application Development

AI Mock Interview Platform is a web-based system, The platform will be developed using React, Next.js, Tailwind CSS, and TypeScript for the frontend, with Firebase services (Authentication, Firestore Database, Hosting, and Cloud Functions) powering the backend. Frontend components must be modular, reusable, and responsive to ensure smooth user interaction across devices. Backend services should follow RESTful principles, be welldocumented, and allow seamless integration with AI modules for interview evaluation and third-party services.

2.2.9 Documentation

Prepare comprehensive user documentation, API documentation, and developer guides to ensure smooth onboarding, maintenance, and future system enhancements. Documentation should include step-by-step instructions, screenshots, and code examples where applicable. Maintaining updated documentation will support knowledge transfer, facilitate debugging, and accelerate new feature integration.

2.2.10 Testing and Quality Assurance

Conduct rigorous testing, including unit tests, integration tests, and system tests to ensure the platform is stable, efficient, and error-free. Automated testing frameworks should complement manual testing to cover both edge cases and typical workflows. Test reports should document functionality, performance, security, and usability findings, providing actionable insights for continuous improvement.

2.2.11 Regular Updates and Maintenance

An ongoing maintenance and upgrade plan will be established for the AI Mock Interview Platform to refine AI-driven interview models, improve performance, and ensure continued system security. This plan will include periodic system audits, usability testing with students and recruiters, and performance benchmarking to identify areas for improvement. Regular updates will maintain compatibility with evolving technologies, incorporate user feedback, and enhance the accuracy of AI-generated feedback and recruiter dashboards. These deliverables and development requirements collectively ensure the successful development and long-term evolution of the AI Interview Platform, aligning closely with its vision of supporting interview readiness, improving communication clarity, and enabling smarter recruiter–candidate matching. Furthermore, they establish a structured foundation for integrating future AI-driven features, scaling the system as the user base expands, and maintaining consistency, reliability, and adaptability throughout its lifecycle.

2.3 Current System Limitations

Currently, there is no centralized platform that enables students and fresh graduates to effectively prepare for interviews in a structured and academically focused way [8]. Most individuals rely on static tutorials, scattered online resources, and general job portals, which are not tailored to interview readiness or communication skill development [9]. As a result, students often struggle to build confidence, improve clarity in responses, and present their skills in a professional manner, leading to incomplete preparation and reduced chances of success when applying for internships, research roles, or entry-level jobs [10].

Users also lack guided tools that help them practice lifelike interview scenarios, receive constructive feedback, and understand how to communicate their academic and professional experiences effectively [11]. Without a dedicated system to simulate interviews or provide structured feedback, many students experience confusion, low confidence, and difficulty presenting themselves to recruiters.

The following table 2.1 summarizes the limitations of existing interview preparation and professional platforms, emphasizing the need for an integrated, user-friendly solution like the AI Mock Interview Platform, which focuses on AI-driven mock interviews, feedback generation, and recruiter–student engagement. These gaps highlight the absence of a unified system that streamlines student development workflows while supporting personalized career growth. The analysis further reinforces the importance of a platform that bridges academic preparation with real-world recruitment opportunities, as given in Table 2.1.

Table 2.1 Current System Limitations

Platform	Limitations
LinkedIn	Not designed for students or fresh graduates; lacks structured interview practice modules; mixes corporate and academic content; limited support for guided interview preparation.
Academia.edu	Focuses mainly on research sharing; it lacks mock interview features, skill evaluation tools, and recruiter-focused dashboards.
Google Sites / Wix	Requires design and layout skills; no guided interview structure; no built-in recruiter interaction or AI-driven feedback features [10].
Traditional CVs (PDF/DOC)	Static and difficult to update; cannot simulate interviews; limited visibility; lacks centralization and professional readiness evaluation [11].
ORCID	Primarily ID and publication tracking; not an interview preparation platform; lacks sections for skills, projects, certifications, or recruiter access.

2.4 Summary

The AI Mock Interview Platform project addresses the challenges faced by students and fresh graduates in preparing for interviews by offering a centralized, structured, and user-friendly system. Existing tools like LinkedIn, Academia.edu, and traditional CVs rely on static templates and scattered information, making it difficult for users to build confidence, practice interviews, and effectively showcase their skills. PrepWise overcomes these limitations by enabling students to engage in lifelike AI-assisted mock interviews, receive constructive feedback, and connect directly with recruiters through a dedicated dashboard. Built using modern technologies such as React, Next.js, Tailwind CSS, TypeScript, and Firebase, the AI Interview Platform ensures secure authentication, smooth navigation, and scalable performance. By integrating AI-driven feedback and recruiter engagement features, the platform bridges the gap between academic preparation and real-world recruitment opportunities, supporting students in their career growth while providing recruiters with reliable insights into candidate readiness.

Chapter 3
Requirement Analysis

3 Requirement Analysis

3.1 Use Case Diagrams

In this Use case diagram, we show the behavior of our system. This enables you to visualize the different types of roles in a system and how those roles interact with the system [12].

3.1.1 Authentication Module

Figure 3.1 illustrates the interactions between a User and Firebase Auth within the Authentication Module. The User performs core authentication tasks such as Register/Sign Up**,** Login/Sign In, and Logout**.** Concurrently, Firebase Auth verifies credentials and manages secure sessions through processes like Sending Credentials to Firebase**,** Authentication via Firebase**,** and redirecting to Dashboard. This diagram highlights how user input flows to Firebase for validation, ensuring secure access and session management in the PrepWise platform.[7]

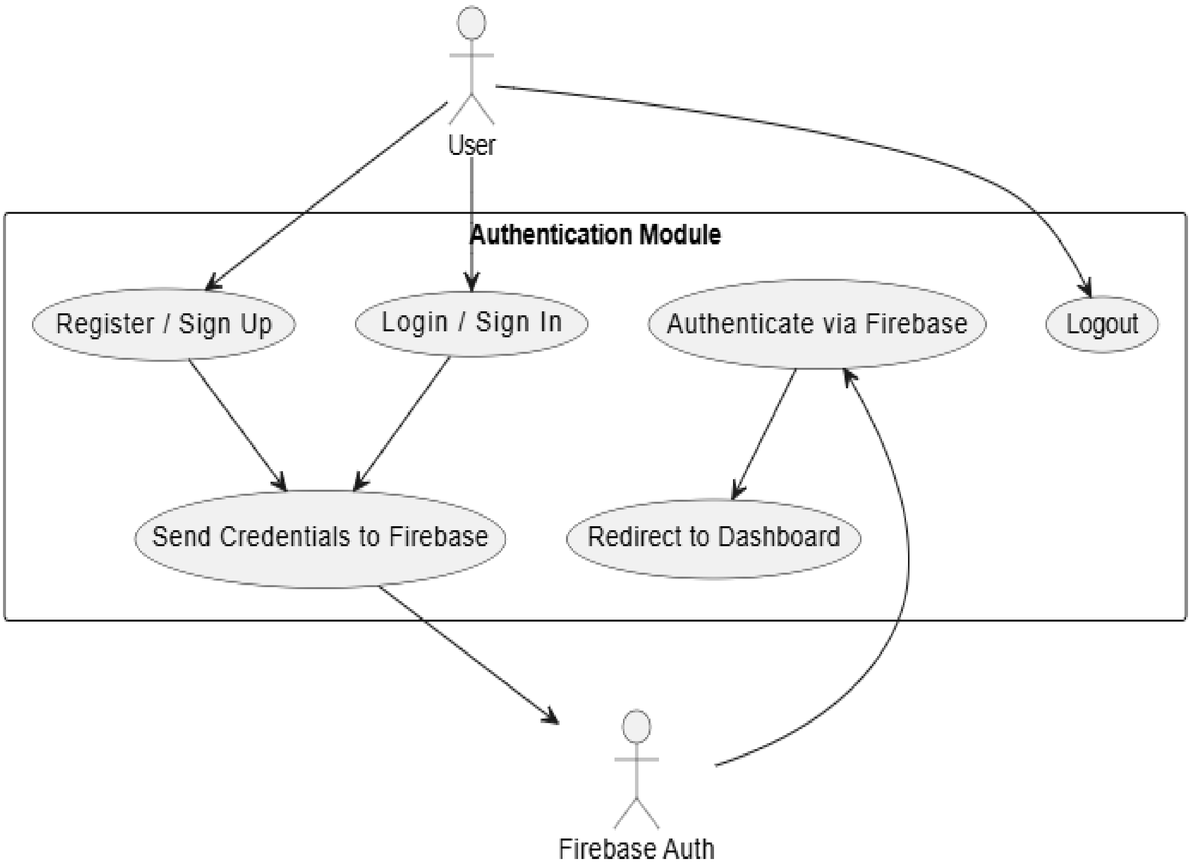

Figure 3.1 Authentication Module

3.1.2 Interview Generation Module

Figure 3.2 illustrates the interactions within the Interview Generation Module [8]. The User provides inputs such as Role & Level**,** Interview Type, and Technologies. These inputs trigger the Generate Custom Interview use case, supported by Vapi Agent and Gemini AI**.**

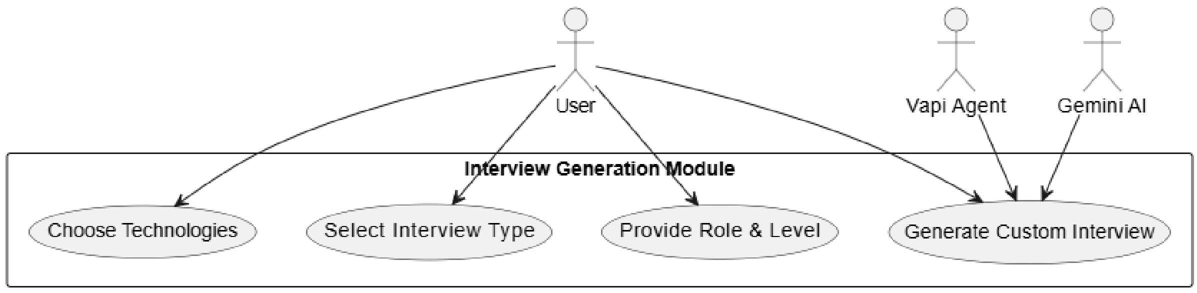

Figure 3.2 Interview Generation Module

3.1.3 Dashboard Module

Figure 3.3 illustrates the interactions within the Dashboard Module. The User can perform two primary actions: View Existing Interviews and Access New Interviews. These use cases highlight the dashboard's role as a central hub, enabling candidates to manage past interview sessions and initiate new ones. The diagram emphasizes how the dashboard streamlines navigation and supports efficient interview management within the PrepWise platform.

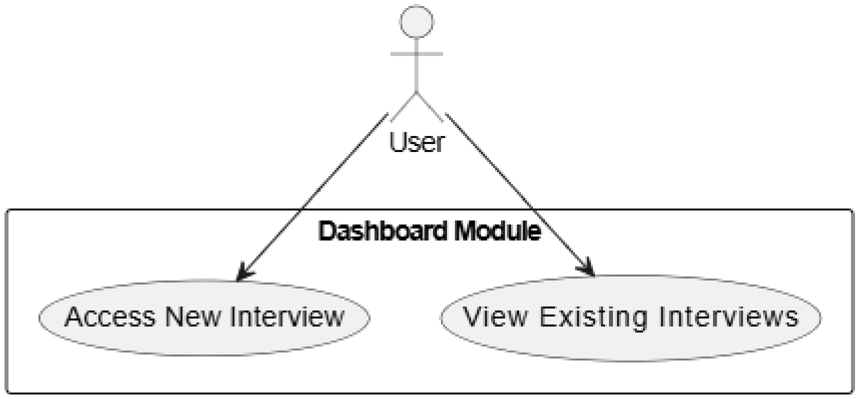

Figure 3.3 Dashboard Module

3.1.4 Take Interview Module

Figure 3.4 illustrates the interactions within the Take Interview Module. The **User** engages in the interview process by Answering Questions and completing the Interview Session**.** Meanwhile, the Vapi Interviewer initiates the session through the Start Mock Interview use case. This diagram highlights how candidate participation and AI-driven interviewing combine to simulate realistic interview experiences within the PrepWise platform.

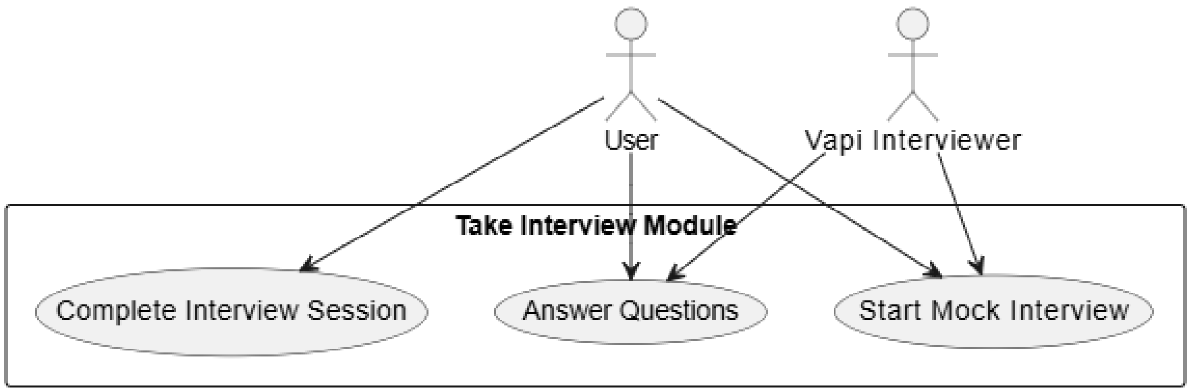

Figure 3.4 Take Interview Module

3.1.5 Feedback Module

Figure 3.4illustrates the interactions within the Feedback Module. The User engages with the system by generating and viewing feedback reports after completing interviews. The Gemini AI supports this process by analyzing responses and producing structured feedback. This diagram highlights how AI assistance and user interaction combine to deliver actionable insights, enabling candidates to identify strengths and areas for improvement within the PrepWise platform.[15]

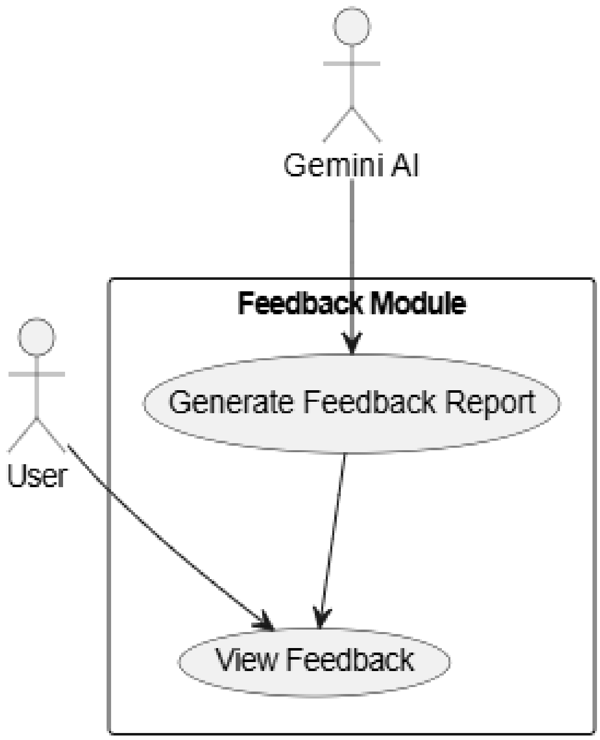

Figure 3.5 Feedback Module

3.2 Use Case Description:

3.2.1 Authentication Module

Table 3.1 defines how users securely access the PrepWise platform through the Authentication Module. It outlines the steps involved in sign-up, sign-in, session validation, and logout**,** ensuring clarity in user interactions, system responsibilities, and Firebase-supported authentication. This description forms the foundation for secure access and controlled navigation within the platform.

Table 3.1 Authentication Module

Use Case ID	**UC-01**
Use Case Name	Authentication and Session Management
Actors	**Primary:** User **Secondary:** Firebase Auth (for credential verification and token management)
Description	Users register, log in, and manage sessions through Firebase authentication. The system validates credentials, generates secure tokens, and redirects authenticated users to the dashboard.
Trigger	User initiates sign-up, sign-in, or logout.
Preconditions	PRE-1: User must provide valid credentials. PRE-2: Firebase Auth service must be available.
Postconditions	POST-1: User session is created, validated, or terminated. POST-2: Authenticated users are redirected to the dashboard.
Normal Flow	1. User initiates sign-up or sign-in. 2. System collects credentials and sends them to Firebase Auth. 3. Firebase verifies credentials and generates an ID token. 4. System creates a secure session cookie. 5. User is redirected to the dashboard. 6. On logout, the session is cleared and user is redirected to sign-in page.
Alternative Flows	1. Invalid credentials → system prompts user to re-enter. 2. Session expired → user must re-authenticate.
Exceptions	E1: Firebase service unavailable → system displays error message. E2: Missing required fields → system prompts user to complete input.
Business Rules	BR-1: Only authenticated users can access protected routes. BR-2: Session cookies expire after 7 days. BR-3: Secure (HTTPS) connections are mandatory in production.
Assumptions	1. Users prefer simple email/password authentication. 2. Firebase Auth ensures secure credential management. 3. System redirects authenticated users automatically to the dashboard.

3.2.2 Interview Generate

Table 3.2 explains how the Interview Generation Module processes user inputs to create customized interview sessions. It highlights the interaction between the User and Vapi Agent and Gemini AI as they handle role, level, type, and technology selections. The description outlines how tailored interview questions are generated, ensuring lifelike simulation and relevance. It also emphasizes how AI-driven customization evolves with user preferences, enabling continuous improvement in interview preparation.

Table 3.2 Interview Generation Module

Use Case ID	UC-02
Use Case Name	Interview Generation
Actors	**Primary:** User **Secondary:** Vapi Agent, Gemini AI (for customization and lifelike simulation)
Description	Users generate customized mock interviews by providing inputs such as role & level, interview type, and technologies. The system, supported by AI agents, creates tailored interview sessions aligned with candidate needs.
Trigger	User initiates interview generation by selecting parameters (role, type, technologies).
Preconditions	PRE-1: User must be logged in. PRE-2: System must have AI agents (Vapi, Gemini) available.
Postconditions	POST-1: A customized interview session is generated. POST-2: Interview is saved and accessible via the dashboard.
Normal Flow	1. User selects role & level. 2. User chooses interview type. 3. User specifies technologies. 4. System collects inputs and triggers interview generation. 5. Vapi Agent and Gemini AI process inputs. 6. AI generates customized interview questions. 7. System saves the generated interview and displays a preview.
Alternative Flows	1.1 User edits parameters after generation → system regenerates interview. 1.2 AI suggests default questions if inputs are incomplete. 1.3 User cancels generation → system discards inputs.
Exceptions	E1: Missing required inputs → system prompts user to complete. E2: AI service unavailable → system generates basic interview template.
Business Rules	BR-1: Each interview must be linked to a user profile. BR-2: AI agents must validate question relevance before saving. BR-3: Generated interviews can be previewed before saving.
Assumptions	1. Users prefer AI-assisted customization. 2. AI enhances realism but does not replace user control. 3. Users may refine interview parameters later.

3.2.3 Dashboard Module

Table 3.3 explains how users interact with the Dashboard Module to manage interview sessions within the PrepWise platform. It highlights the interaction between the User and the system as they access new interviews or review existing ones. The description also outlines how the dashboard serves as a central hub, ensuring efficient navigation, streamlined access, and accurate management of interview records.

Table 3.3 Dashboard

Use Case ID	UC-03
Use Case Name	Dashboard Access and Interview Management
Actors	**Primary:** User
Description	Users interact with the dashboard to view existing interviews and access newly generated ones. The dashboard provides centralized navigation and ensures efficient interview management.
Trigger	User logs in and navigates to the dashboard.
Preconditions	PRE-1: User must be authenticated. PRE-2: System must have stored interview records.
Postconditions	POST-1: User successfully views past interviews. POST-2: User accesses newly generated interviews.
Normal Flow	1. User logs in and opens the dashboard. 2. System displays available interviews. 3. User selects either "View Existing Interviews" or "Access New Interview." 4. System loads the requested interview session.
Alternative Flows	1.1 User filters interviews by date or type → system updates displayed list. 1.2 User cancels navigation → system returns to dashboard home.
Exceptions	E1: No interviews available → system informs user and suggests generating a new interview. E2: Database error → system displays error message.
Business Rules	BR-1: Only authenticated users can access dashboard features. BR-2: Dashboard must display both past and new interviews. BR-3: Interview records update dynamically as new sessions are generated.
Assumptions	1. Users prefer centralized access to interviews. 2. Dashboard serves as the main navigation hub. 3. Users may revisit past interviews for preparation.

3.2.4 Take Interview Module

Table 3.4 explains how users interact with the Take Interview Module to simulate real interview sessions within the PrepWise platform. It highlights the interaction between the User and the Vapi Interviewer, showing how mock interviews are initiated, answered, and completed. The description emphasizes how AI-driven interviewing creates lifelike experiences, enabling candidates to practice communication, confidence, and problem-solving skills.

Table 3.4 Take Interview

Use Case ID	UC-04
Use Case Name	Take Mock Interview
Actors	**Primary:** User **Secondary:** Vapi Interviewer (AI)
Description	Users participate in AI-driven mock interviews by answering questions and completing interview sessions. The Vapi Interviewer initiates the

	session and simulates realistic interview dynamics.
Trigger	User selects “Start Interview” from the dashboard.
Preconditions	PRE-1: User must be logged in. PRE-2: AI interviewer service must be available.
Postconditions	POST-1: Interview session is completed and saved. POST-2: User responses are recorded for feedback analysis.
Normal Flow	1. User selects “Take Interview” option. 2. Vapi Interviewer initiates the mock interview. 3. User answers AI-generated questions. 4. System records responses. 5. User completes the interview session.
Alternative Flows	1.1 User pauses interview → system saves progress. 1.2 User exits early → system marks interview as incomplete.
Exceptions	E1: AI interviewer unavailable → system prompts user to retry later. E2: Connection lost → system saves partial responses.
Business Rules	BR-1: Each interview session must be linked to a user profile. BR-2: User responses are stored securely for feedback generation. BR-3: Interview sessions cannot be edited once completed.
Assumptions	1. Users prefer lifelike AI-driven interviews. 2. Vapi Interviewer ensures realistic simulation. 3. Users may retake interviews for improvement.

3.3 Functional Requirements

Functional requirements specify what the AI Interview Platform must do to fulfill user needs and expectations. They define the exact features, operations, and behaviors the system should provide, such as interview generation, real-time simulation, AI-based feedback, and recruiter job modules. Their importance lies in establishing a shared understanding among developers, users, and stakeholders, ensuring that the system’s capabilities are fully aligned with project goals. By identifying these functions early, functional requirements help avoid misunderstandings, minimize costly revisions, improve planning accuracy, and support thorough testing. Ultimately, they guide the overall development process and ensure the final product delivers meaningful, reliable, and user-centered value.

The functional requirements of the AI Interview Platform encompass a wide range of features designed to support candidates and recruiters through structured workflows and intelligent tools. Each function contributes to improving interview preparation, recruiter efficiency, and overall engagement. The key functional requirements include:

3.3.1 User Authentication and Authorization

The system shall provide secure login and registration using Firebase Auth. Role-based access ensures candidates and recruiters only access relevant features. Password recovery and session management improve usability and reliability..

3.3.2 Dashboard Management

The platform shall allow users to view existing interviews and access newly generated ones**.** The dashboard serves as a central hub for navigation, ensuring efficient interview management and streamlined access.

3.3.3 Interview Generation

The system shall enable users to generate customized interviews by selecting role, level, interview type, and technologies**.** Vapi Agent, Gemini AI processes inputs to create lifelike interview sessions tailored to candidate needs.

3.3.4 Real-Time Mock Interview

The platform shall simulate interviews in real time. The Vapi Interviewer initiates sessions, while candidates answer AI-generated questions. Responses are recorded for later feedback analysis.

3.3.5 Feedback and Retake Module

The system shall generate structured feedback reports using Gemini AI, highlighting strengths and weaknesses. Candidates can retake interviews multiple times to improve performance.

3.4 Nonfunctional Requirements

Nonfunctional requirements define the quality attributes and performance standards that determine how the AI Interview Platform operates. They ensure reliability, efficiency, and positive user experience.

3.4.1 Performance

The platform shall ensure fast loading times across all core features, including dashboard access, interview generation, real-time mock interviews, feedback reports, and recruiter job modules. AI-driven processes such as question generation, response recording, ranking, and analytics must execute efficiently without noticeable delays. The system should be optimized to handle high traffic volumes and hundreds of concurrent interview sessions without performance degradation. Consistent performance is essential to maintaining user trust, recruiter engagement, and candidate confidence during interview preparation.

3.4.2 Security

The system shall safeguard candidate and recruiter data through strong encryption, secure authentication mechanisms, and strict access control policies. Sensitive information such as interview responses, recruiter job postings, and feedback reports must be securely stored and transmitted. The platform will follow data protection principles similar to GDPR to ensure user privacy, transparency, and compliance. Regular security checks, validations, and monitoring will further reduce the risk of unauthorized access, data breaches, or misuse, ensuring trust and reliability in the AI Interview Platform.

3.4.3 Availability

The platform should maintain high availability to ensure candidates and recruiters can access services at all times. Users should be able to generate interviews, take mock sessions, view feedback reports, and manage recruiter job postings without interruption. Minimal downtime and reliable server infrastructure are critical to supporting global usage. This requirement ensures continuity of interview preparation, recruiter engagement, and overall platform reliability.

3.4.4 Ease of Use

The platform's interface should be intuitive and user-friendly for candidates and recruiters with varying levels of technical expertise. Navigation must be simple, consistent, and well-structured, with clear labels and visually clean layouts. The system will support responsive design for all devices and consider accessibility standards to accommodate diverse users. Future enhancements may include multilingual support to further broaden usability and reach, ensuring that both candidates and recruiters can efficiently interact with the AI Interview Platform.

Chapter 4
Design and Architecture

4 Design and Architecture

4.1 System Architecture

Figure 4.1 explains how AI Interview Platfofrm operates by describing its core components, how they are organized, and the way they interact with each other to ensure smooth functionality throughout the platform.

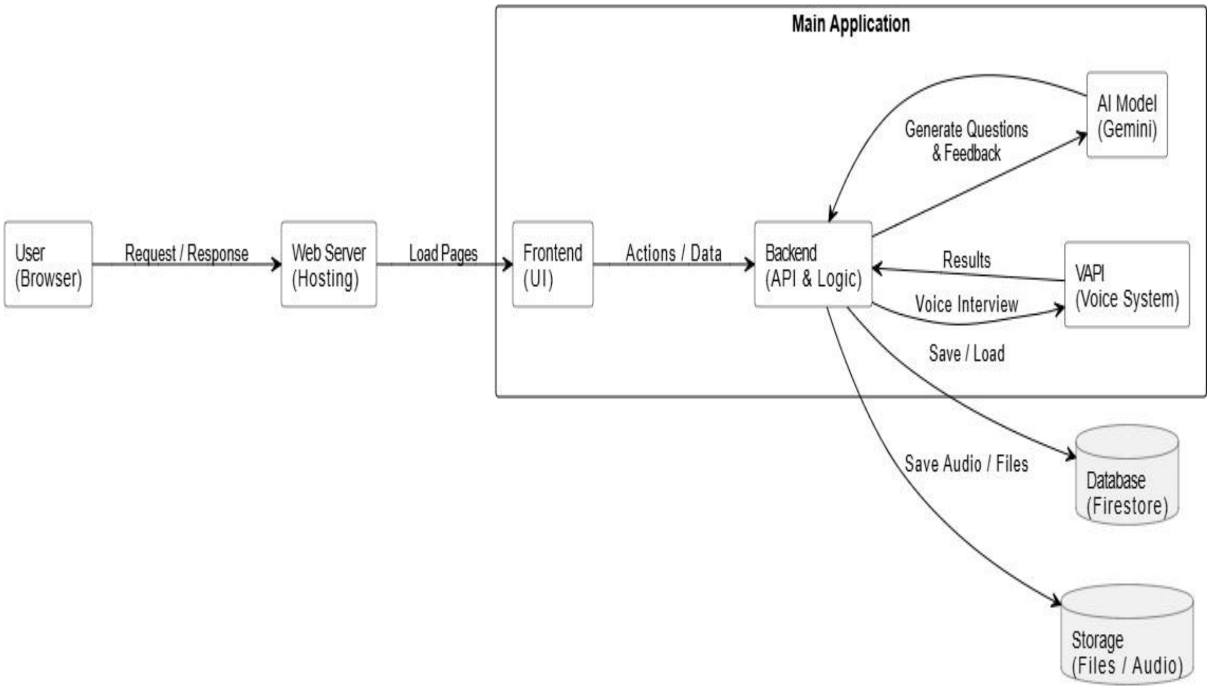

Figure 4.1 AI Interview Platform Architecture

The architecture of the AI Interview Platform follows a modern client–server model designed to be modular, scalable, and easy to maintain. The client side is developed using Next.js, React, Tailwind CSS, and TypeScript, providing fast, responsive, and intuitive interfaces for authentication, dashboard navigation, interview generation, real-time mock interviews, and recruiter job modules.

The backend is powered by Firebase services and integrated APIs, where the business logic manages core functionalities such as user authentication, interview generation, mock interview handling, feedback processing, and recruiter–candidate interactions. This ensures a clean separation between the interface and application logic.

The system uses Firestore as the primary database for storing user profiles, interview records, recruiter data, and feedback reports. File and audio storage is handled through Firebase Storage, ensuring secure and reliable access to uploaded resumes, certificates, and recorded interview sessions. The platform integrates Gemini AI for generating interview questions and structured feedback, while VAPI enables real-time voice interviews and lifelike simulation.

Role-Based Access Control (RBAC) ensures that candidates and recruiters can only access features relevant to their roles, while secure API communication and encrypted storage protect sensitive user data.

The AI Interview Platform's modular architecture supports future enhancements and scalability, allowing additional modules such as mobile applications, advanced AI features, or multi-language support to be integrated smoothly. This structure ensures efficient performance, maintainability, and a seamless user experience across all components of the platform.

4.2 Data Representation

In the AI Interview Platform, all candidate, recruiter, and interview-related data are stored within structured cloud databases to ensure fast, consistent, and reliable access to information. The system utilizes Firestore with well-defined collections and documents to systematically organize user profiles, interview records, recruiter job postings, feedback reports, and audio files while maintaining strong relationships among these entities. Unlike simple unstructured storage systems, this document-based approach supports indexing, querying, and constraint enforcement, which are essential for maintaining data integrity and reducing redundancy.

The structured data model allows the AI Interview Platform to support advanced querying mechanisms required for AI-based interview generation, response tracking, feedback analysis, and recruiter talent discovery. Efficient data encoding and decoding facilitate real-time updates, seamless synchronization of interview sessions, and responsive recruiter searches. Furthermore, the clean and normalized design ensures smooth integration with machine learning models by providing query-optimized datasets. As a result, this approach significantly enhances system accuracy, scalability, and performance, ultimately delivering a faster, more intelligent, and user-centric experience across the platform.

4.3 Process Flow

Figure 4.2 The process flow representation for the AI Interview Platform illustrates the complete journey of candidates and recruiters as they interact with the system. Candidates begin by registering and authenticating through the Next.js frontend with Firebase Authentication. Once logged in, they build and update their interview profiles, specifying skills, education, and experience. Candidates then generate AI-driven mock interviews by selecting role, level, and technology stack, after which Gemini AI produces tailored interview questions. They can participate in real-time sessions, receive structured feedback, and retake interviews to improve confidence and performance for real job opportunities.

Recruiters follow a parallel and simplified process where they register, authenticate, and access their dashboards. Through Firestore queries and AI-based ranking, recruiters can search candidate profiles, filter by skills or interview performance, and post job opportunities. They view ranked candidate profiles and initiate contact with suitable applicants. The diagram demonstrates how both user types interact smoothly with the platform, highlighting an intuitive workflow, modular design, and clear separation of roles. Overall, it captures how AI Interview Platform ensures efficiency, intelligent evaluation, and seamless communication between candidates and recruiters, ultimately creating a user-friendly environment that supports interview preparation and early-career hiring.

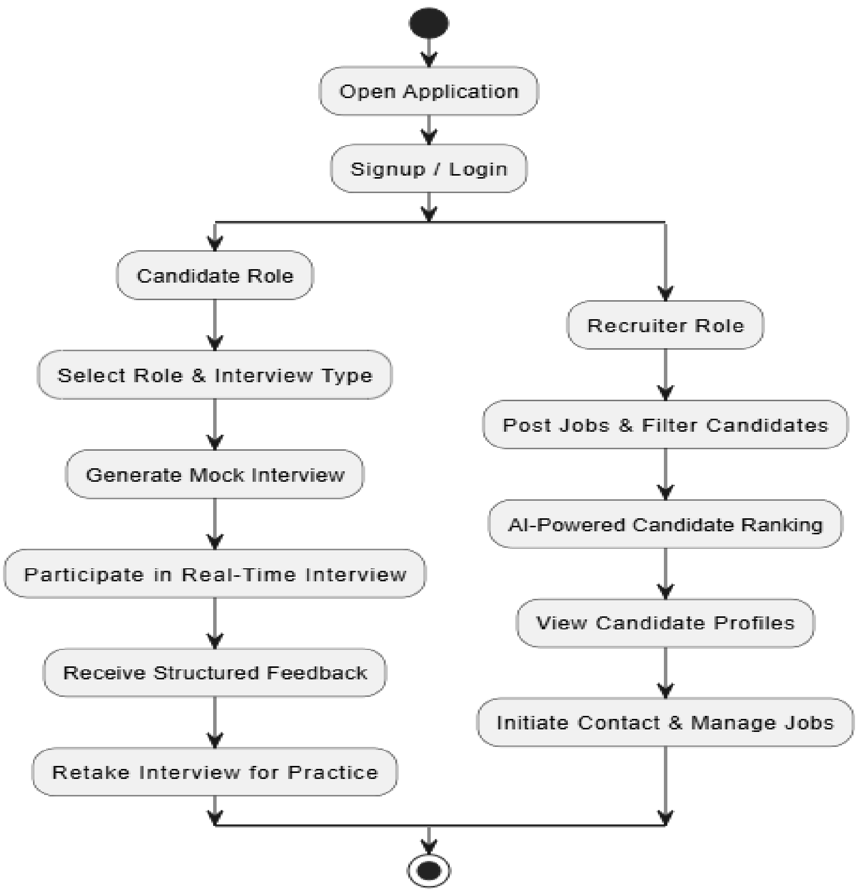

Figure 4.2 AIP Process Flow Diagram

1. **User Registration and Login**

   Users begin by creating an account and logging into the AI Interview Platform. After successful authentication, they are redirected to their personalized dashboard based on their role candidate or recruiter.

2. **For Candidates**

   Build Profile: Candidates can create and update their profiles by adding education, skills, experience, and preferred job roles.

   Generate Mock Interview: They can select interview type, role, and technology to generate AI-driven mock interviews using Gemini AI.

   Participate in Real-Time Interview: Candidates interact with the VAPI voice system to simulate lifelike interview sessions.

   Receive Feedback: After completion, candidates receive structured feedback reports highlighting strengths and improvement areas.

   Retake Interviews: They can retake interviews multiple times to enhance confidence and performance.

3. **For Recruiters**

   Recruiter Dashboard: Recruiters access a dedicated dashboard designed for posting jobs, filtering candidates, and evaluating interview readiness.

   Post Jobs and Filter Candidates: Recruiters can create job listings and filter candidates based on education, skills, and AI-driven interview performance metrics.

   View Candidate Profiles: Recruiters can view detailed candidate profiles, including structured feedback reports, interview readiness scores, and communication clarity indicators.

   AI-Powered Ranking: The system automatically ranks candidates using AI algorithms that consider skill match, profile completeness, and interview results.

   Contact Candidates: Recruiters can send direct messages or interview invitations through the platform to initiate hiring processes.

4.4 Sequence Diagram

4.4.1 User Registration and Email Verification

Figure 4.3 A new user fills out the registration form in the Next.js frontend, which Firebase Authentication validates and stores as an inactive account in Firestore. The backend then sends a verification email through Firebase Email Service to ensure that the user owns the provided email address. Once the user clicks the verification link, a verification request is sent back to Firebase Authentication. The system validates the token, activates the account, and updates the user's status in Firestore. Finally, the user is notified that their account has been successfully verified.

This process ensures secure and authenticated onboarding while preventing unauthorized access and reducing the risk of fake or duplicate accounts. Additionally, the registration workflow is designed to handle errors gracefully, such as expired tokens or invalid email formats, providing clear feedback to users and maintaining a smooth onboarding experience.

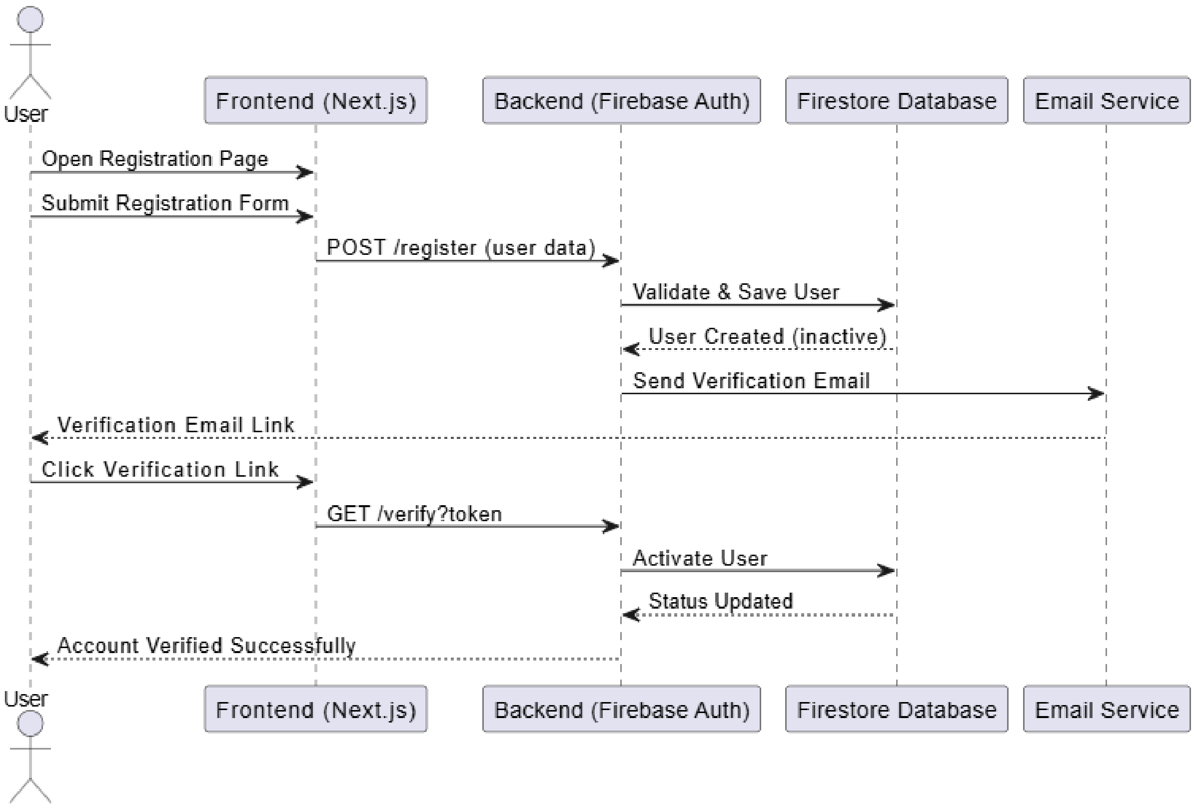

Figure 4.3 User Registration and Email Verification

4.4.2 Login and Role-Based Dashboard

Figure 4.4 llustrates how users log into the AI Interview Platform using a responsive login page built with Next.js and Firebase Authentication. The entered credentials are submitted to Firebase, where authentication and role verification (candidate or recruiter) take place. If credentials are valid, Firebase redirects the user to the appropriate dashboard view based on their role. Incorrect credentials return an error message back to the frontend. Role-based routing ensures that each user accesses only the features intended for them. This maintains security and functionality separation.

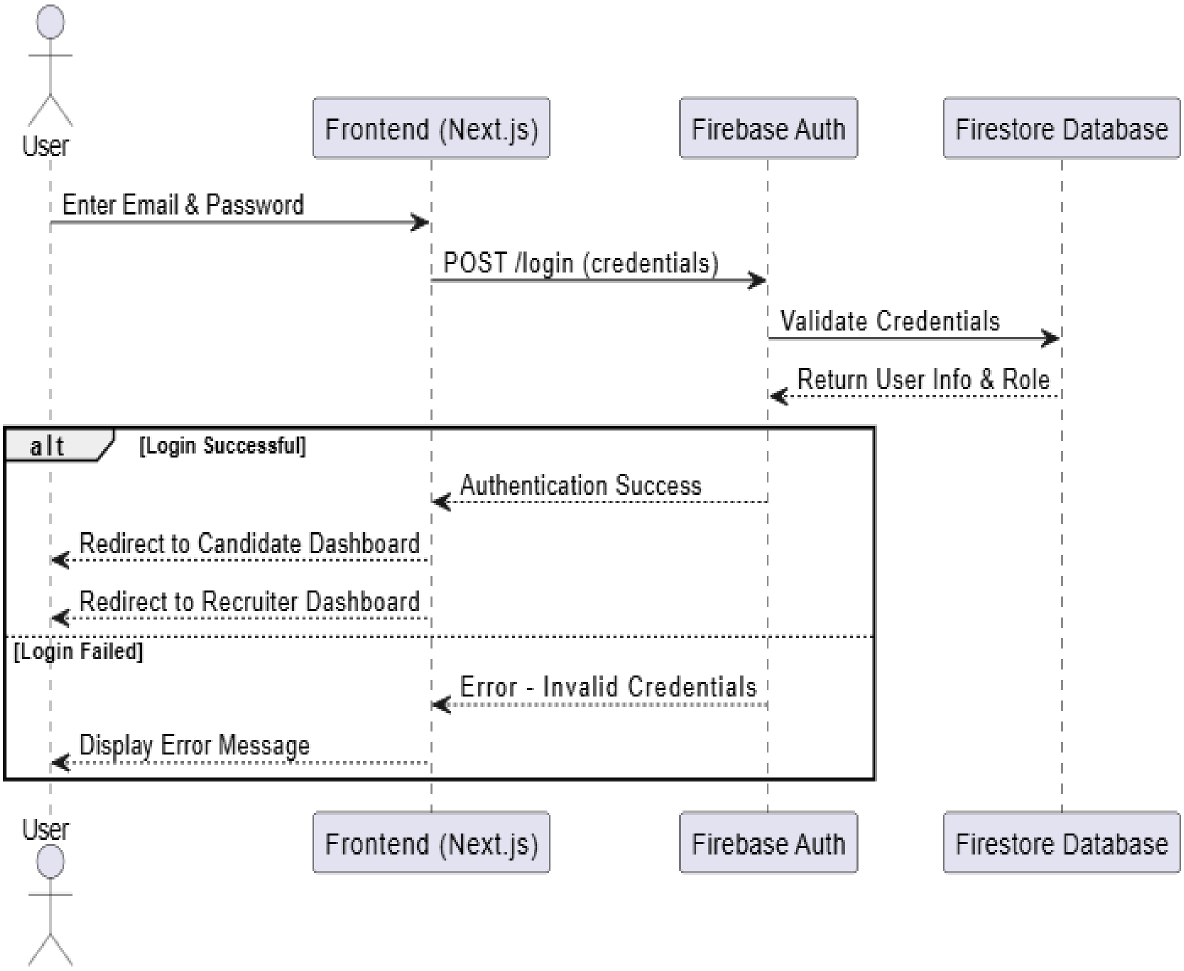

Figure 4.4 Login and Role-Based Dashboard

4.4.3 AI Interview Generation

Figure 4.5 The sequence diagram illustrates how a candidate generates an interview session within the AI Interview Platform. The process begins when the candidate selects role, level, interview type, and technology stack through the Next.js frontend. The frontend sends these parameters to the Firebase backend, which manages the business logic. The backend then communicates with Gemini AI, which generates a tailored set of interview questions based on the candidate's profile and selected parameters.

Once the questions are generated, the backend stores the interview session details in Firestore Database to ensure persistence and future retrieval. A confirmation is returned to the frontend, which delivers the generated interview questions to the candidate. This workflow ensures that interview sessions are personalized, securely stored, and immediately available for practice or real-time execution.

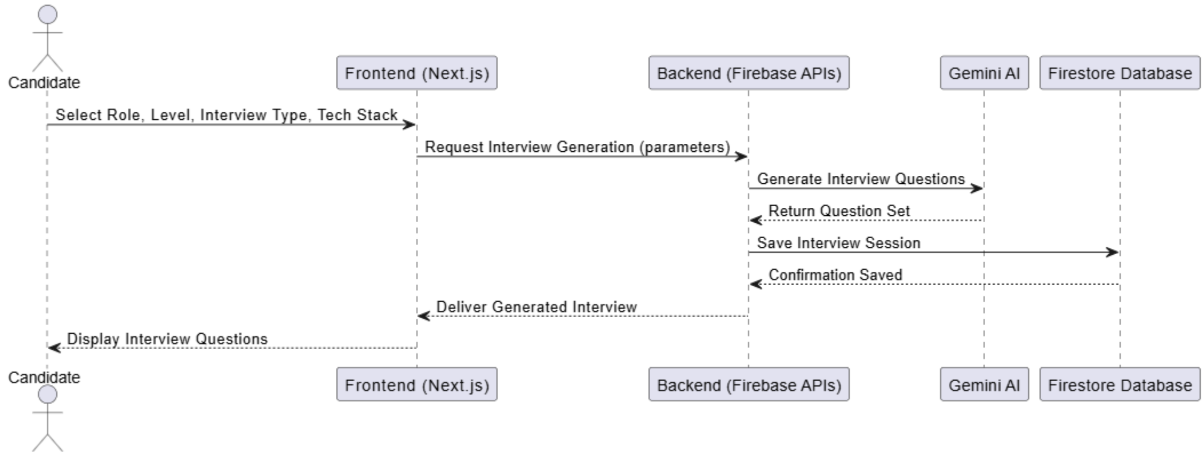

Figure 4.5 AI Interview Generation

4.4.4 Feedback Generation

Figure 4.6 shows The AI Interview Platform generates structured feedback after a candidate completes an interview session. The candidate finishes the interview through the Next.js frontend, which collects responses in text or voice. These responses are sent to the Firebase Backend, where the business logic manages data flow. The backend forwards the responses to Gemini AI for analysis of communication, confidence, technical knowledge, and problem-solving. Gemini AI returns a structured feedback report with performance scores and improvement suggestions. The backend stores this feedback in Firestore Database for persistence and future retrieval. Finally, the frontend delivers the feedback report to the candidate, enabling continuous improvement through AI-driven insights.

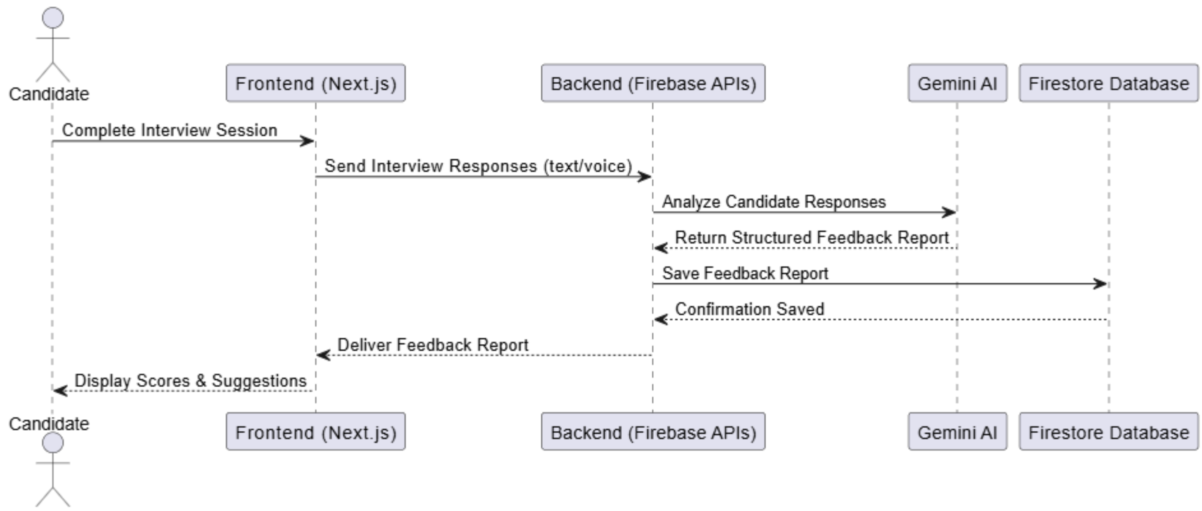

Figure 4.6 Feedback Generation

4.5 Class Diagram

Figure 4.5 The AI Interview Platform provides a comprehensive view of the system's structural design and the relationships between its core modules. At the center of the system is the Candidate class, which represents users preparing for interviews. It contains attributes such as candidate ID, name, email, skills, education, and experience. The Candidate class provides methods to update the profile, participate in interviews, and view feedback reports. This ensures that candidates can continuously refine their information and track their progress.

The Recruiter class models employers or organizations that use the platform to identify talent. Recruiters can post jobs, search for candidates using filters such as skills or education, and contact suitable applicants directly. This class emphasizes the recruiter's role in managing opportunities and interacting with candidates.

The Interview class manages the lifecycle of interview sessions. It stores details such as interview ID, candidate ID, role, technology, and status. Its methods allow the generation of AI-driven questions, conducting real-time sessions, and enabling retakes for practice. This class acts as the bridge between candidates and recruiters, ensuring structured interview management.

The Feedback class captures the results of interviews. It includes attributes such as feedback ID, interview ID, performance scores, and improvement suggestions. Its methods generate structured reports and allow candidates to view their strengths and weaknesses. This class is directly linked to the Interview class, ensuring that every interview session produces actionable insights.

The AIEngine Gemini AI class is responsible for intelligent analysis. It processes candidate responses, evaluates communication clarity, confidence, technical knowledge, and problem-solving ability, and recommends improvements. By integrating with both Interview and Feedback classes, it ensures that the platform delivers personalized, AI-driven evaluations.

Finally, the Database Firestore class provides persistence and data management. It stores user profiles, interview sessions, and feedback reports, and supports retrieval operations for recruiters and candidates. This ensures scalability, security, and reliable data handling.

The relationships between these classes highlight the modular and interconnected design of the platform: candidates are linked to interviews, interviews generate feedback, recruiters manage interviews and candidate searches, while the AIEngine and Firestore database provide intelligence and storage. Together, these components create a seamless, secure, and intelligent environment for interview preparation and recruitment.

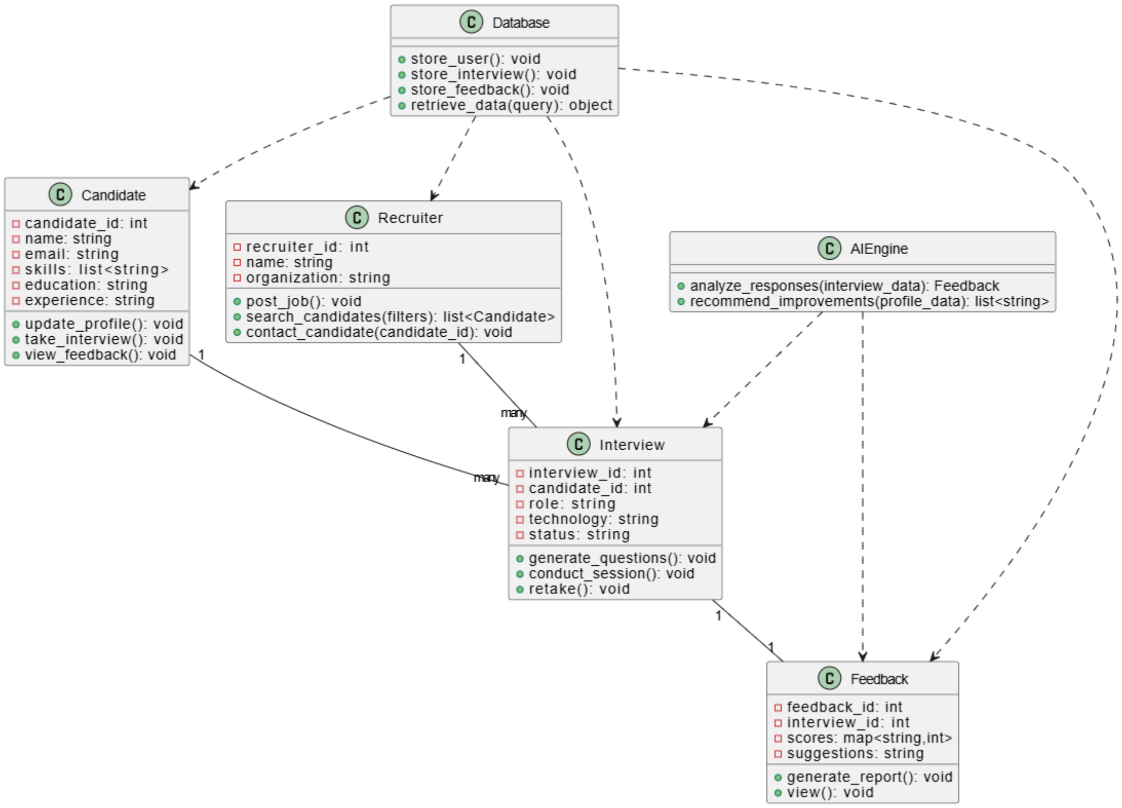

Figure 4.7 AMIP Class Diagram

4.5.1 Components

Candidate Class: Represents users preparing for interviews. It stores personal details, skills, education, and experience, and provides methods to update profiles, take interviews, and view feedback.

Recruiter Class: Models employers who post jobs, filter candidates, and contact suitable applicants. It ensures recruiters can manage opportunities and interact with candidates effectively.

Interview Class: Manages interview sessions, including attributes such as role, technology, and status. It supports generating AI-driven questions, conducting sessions, and enabling retakes for practice.

Feedback Class: Captures interview results, including performance scores and improvement suggestions. It generates structured reports and allows candidates to review their strengths and weaknesses.

AIEngine (Gemini AI) Class: Provides intelligent analysis of candidate responses, evaluates communication and technical skills, and recommends improvements. It integrates closely with Interview and Feedback modules.

Database (Firestore) Class: Ensures pe rsistence and scalability by storing user profiles, interview sessions, and feedback reports. It supports secure retrieval and management of data for both candidates and recruiters.

4.5.2 Relationships

Candidate and Interview: Each candidate can participate in multiple interviews, while every interview is linked to a single candidate. This one-to-many relationship ensures that candidate performance can be tracked across different sessions.

Interview and Feedback: Each interview generates one feedback report, establishing a one-to-one relationship. This guarantees that every interview session produces structured evaluation data.

Recruiter and Interview: Recruiters can manage multiple interviews, either by posting jobs or inviting candidates. This one-to-many relationship highlights the recruiter’s role in overseeing interview processes.

AIEngine and Interview and Feedback: The AIEngine (Gemini AI) interacts with both interviews and feedback. It analyzes candidate responses during interviews and generates feedback reports, ensuring intelligent evaluation and personalized recommendations.

Database and All Classes: The Firestore Database maintains persistence by storing candidate profiles, recruiter data, interview sessions, and feedback reports. It supports retrieval operations, making it central to all relationships.

Chapter 5
Implementation

5 Implementation

In the implementation phase of the AI Interview Platform, we developed core algorithms that ensure smooth, intelligent, and efficient system performance, enabling students, fresh graduates, and recruiters to interact seamlessly. These algorithms power essential features such as AI-driven interview simulations, skill evaluation, recruiter dashboard optimization, and secure data handling, ensuring a reliable and impactful user experience across the platform.

5.1 Algorithm

The core intelligence of the AI Interview Platform relies on optimized algorithms for accuracy and scalability. These algorithms enable lifelike interview simulations, predictive responses, and structured feedback. Integration with VAPI ensures realistic voice-based conversations replicating professional interviews. Efficient API communication supports real-time updates and seamless backend networking. REST standards and asynchronous operations guarantee smooth system interaction. Together, these algorithms deliver reliable performance and impactful user experience across the platform.

1. User Authentication

Function: authenticate_user(username, password)

Input: username (string), password (string)

Output: user (object) or error (string)

Pseudocode:

1. Retrieve user records from the database based on the provided username.
2. If a user with the given username exists:
   a. Compare the provided password with the stored password hash.
   b. If the passwords match, return the user object.
   c. If the passwords do not match, return an error message "Incorrect password".
3. If no user is found, return the error message "User not found".

2. User Profile and Interview Management

Function: manage_profile(user, profile_data)

Input: user (object), profile_data (object containing education, skills, and interview history)

Output: success (Boolean) or error (string)

Pseudocode:

1. Verify that the candidate is authenticated.

2. Validate profile data for required fields (education, skills, interview practice records).
3. If valid: a. Save or update the candidate's profile data in the database. b. Invoke AI module to suggest communication skill improvements and interview readiness tips. c. Store AI suggestions with the profile metadata. d. Return success.
4. If validation fails, return an error indicating missing or incorrect data.

3. Recruiter Search and Ranking

Function: rank_candidates(search_query)

Input: search_query (object with skills, filters, preferences)

Output: ranked_profiles (array of candidate profiles)

Pseudocode:

1. Parse recruiter's search query and extract filters.
2. Retrieve candidate profiles from the database that meet base criteria (education, skills, interview practice records).
3. Compute relevance scores using AI-based ranking algorithms. Features include skill match, profile completeness, interview readiness score, and communication clarity.
4. Apply adaptive learning updates if new recruiter search patterns are detected.
5. Rank candidates based on computed scores.
6. Return ranked_profiles for recruiter dashboard display.

4. AI Interview Generation

Function: generate_interview(user)

Input: user (object)

Output: interview_session (interactive chat)

Pseudocode:

1. Fetch candidate's profile data (education, skills, preferences) from the database.
2. Ask candidate to select interview parameters (role, level, type, tech stack).
3. Generate custom interview questions using AI models.
4. Initiate real-time mock interview session with AI interviewer.
5. Capture candidate's responses for evaluation.
6. Return interview_session ready for feedback and retakes.

5.2 User Interface

The user interface (UI) of AI Interview Platform is designed to be simple, intuitive, and visually appealing.

Figure 5.1 The login page features the PrepWise logo and tagline *"Practice job interviews with AI."* It includes input fields for Email and Password, a prominent Sign In button, and a link for new users to create an account. The design emphasizes clarity and accessibility.

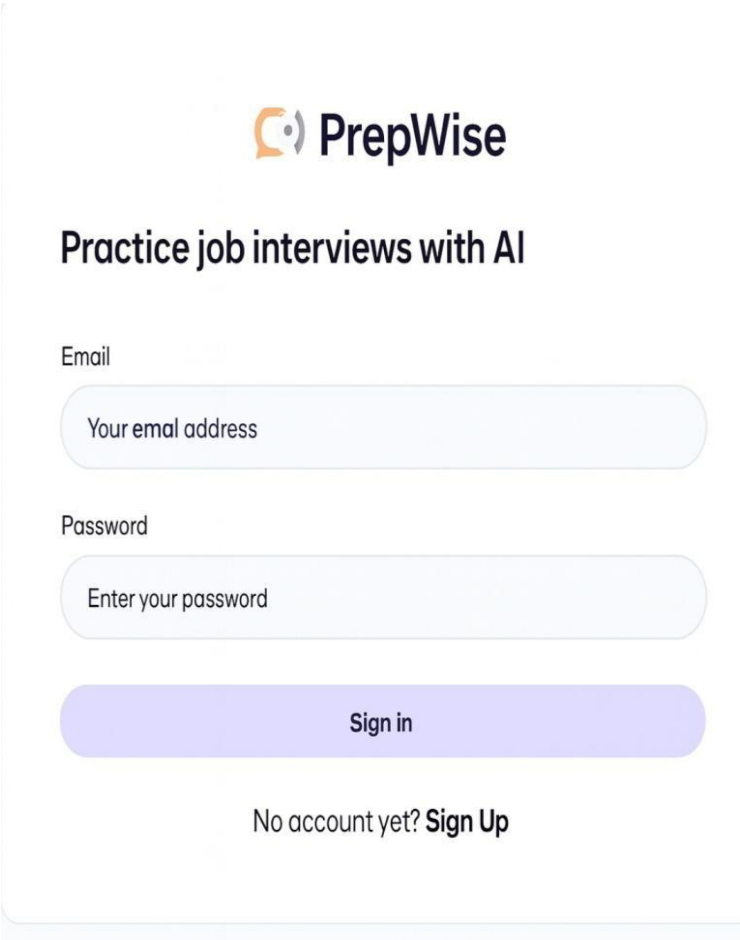

Figure 5.1 Sign-In Page

Figure 5.2 The signup interface allows new users to register by entering their Name, Email, and Password. Create an Account button is displayed, with a link for existing users to sign in. The layout is minimalistic, ensuring ease of use.

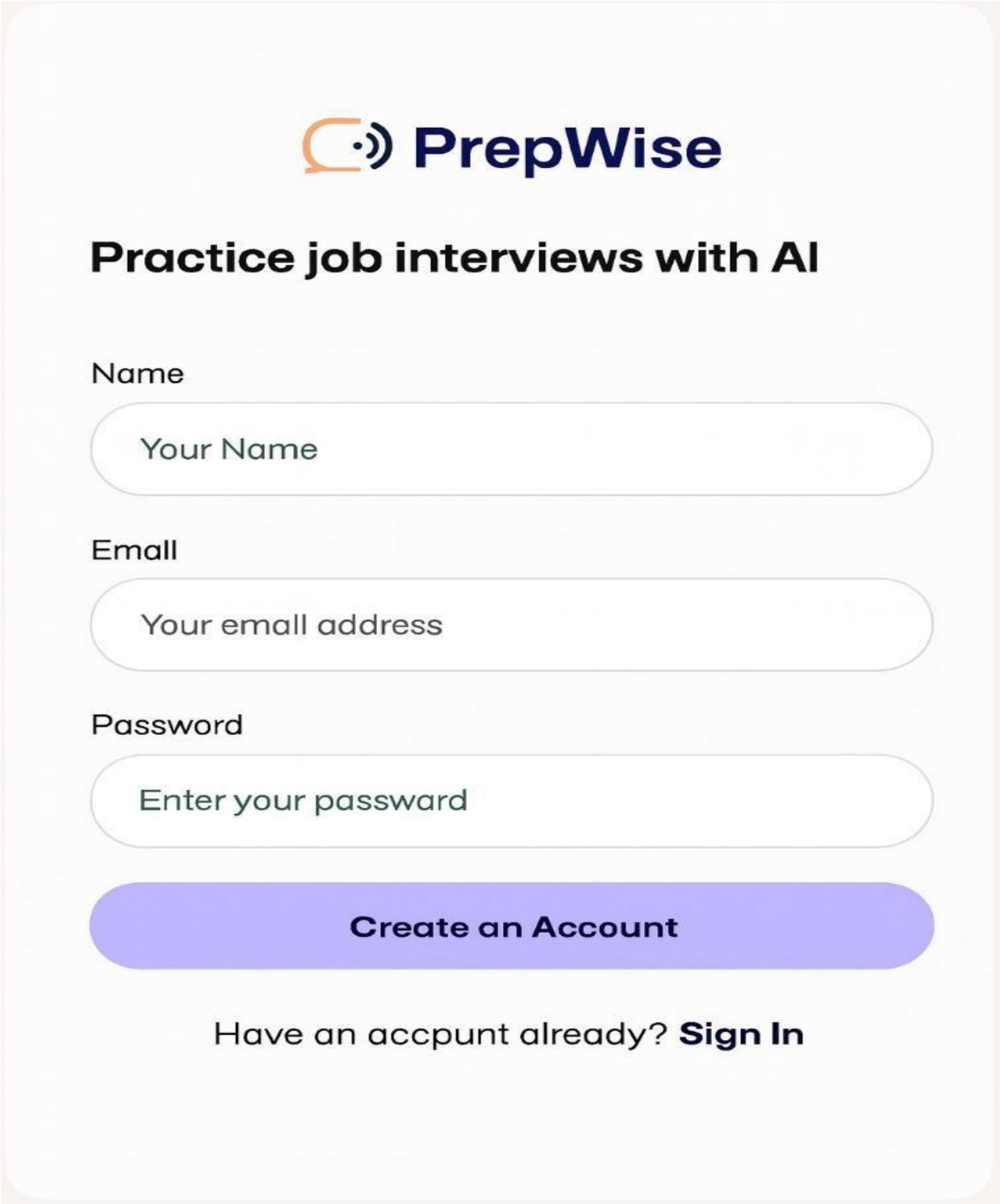

Figure 5.2 Sign-up Page

Figure 5.3 The promotional banner of the AI Interview Platform (PrepWise) highlights the platform's core value: *"Get Interview-Ready with AI-Powered Practice & Feedback."* The banner encourages candidates to practice real interview questions and receive instant feedback. A prominent Start an Interv**iew** button is displayed to motivate immediate engagement.

Figure 5.3 User Dashboard Page

Figure 5.4 This screen presents the AI Interviewer panel alongside the candidate panel, simulating a real-time interview setup. Users can initiate sessions by selecting parameters such as role, level, type, and tech stack. A Call button starts the interactive interview.

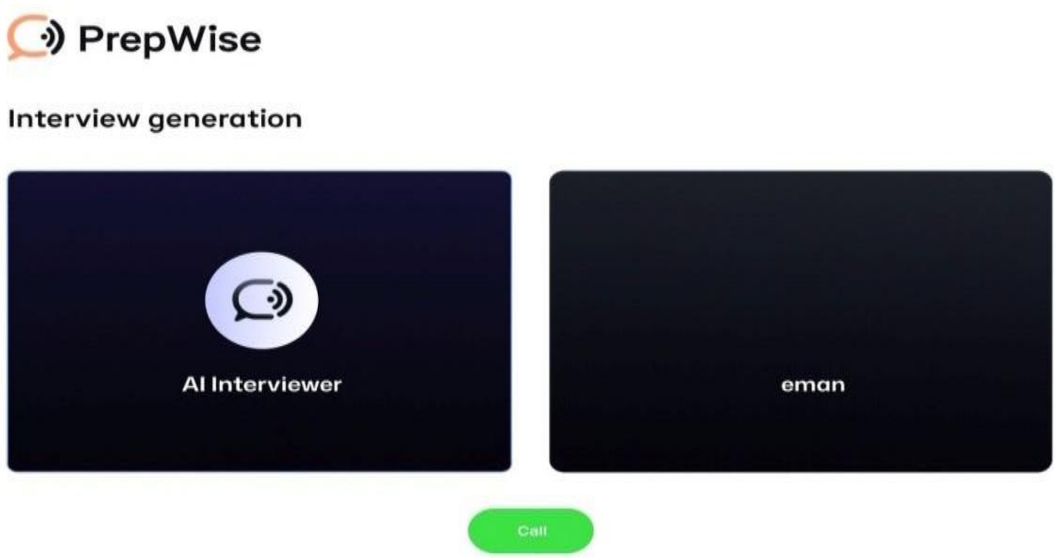

Figure 5.4 Interview Generation Page

Figure 5.5 After completing an interview, the system generates a structured feedback report. It displays an Overall Impression Score and detailed breakdowns of Communication Skills, Technical Knowledge, and Problem-Solving. This helps candidates identify strengths and improvement areas.

Figure 5.5 Feedback Page

5.3 Summary

Chapter Implementation focuses on the practical realization of the AI Interview Platform (PrepWise), detailing how the system's architecture, algorithms, and user interface work together to deliver a seamless mock interview experience. The implementation covers core modules such as authentication (login/signup), dashboard management, AI interview generation, real-time mock interviews, structured feedback system, multiple retakes, and recruiter search and ranking.Intelligent algorithms, including NLP-based question generation, communication skill analysis, confidence scoring, and machine learning models for candidate ranking and adaptive feedback, are integrated into the backend to provide personalized insights and actionable guidance.On the user interface side, the platform employs a responsive and intuitive frontend design using Next.js, React, Tailwind CSS, TypeScript, and Firebase, ensuring smooth navigation across interview generation, real-time sessions, feedback reports, and recruiter dashboards.Together, the backend algorithms and frontend components enable dynamic data handling, real-time updates, and efficient workflows, allowing students, fresh graduates, and recruiters to interact with the system effectively while maintaining consistency, usability, and intelligent automation throughout the platform.

Chapter 6
Testing and Evaluation

6 Testing and Evaluation

Testing in the AI Interview Platform plays a critical role in ensuring that all system components function correctly, efficiently, and cohesively. It verifies that core modules including authentication, dashboard management, AI interview generation, real-time mock interviews, feedback system, multiple retakes, and recruiter search and ranking operate as intended and deliver a smooth, reliable user experience. Testing is conducted across multiple phases of development to identify defects, inconsistencies, and usability issues at an early stage, thereby reducing the risk of system failure after deployment.

From a software engineering perspective, testing is typically performed after implementation to allow independent evaluation of system behavior and performance. This approach ensures that the developed system aligns with defined functional and nonfunctional requirements such as responsiveness, accuracy of AI-generated interviews, and reliability of feedback reports. Through systematic testing and evaluation, PrepWise is validated to reliably support mock interview preparation, structured feedback generation, recruiter interactions, and adaptive learning features, thereby meeting both academic and professional recruitment needs.

6.1 Manual Testing

Manual testing involves the execution of test cases without the use of automated tools, relying instead on direct human interaction with the system. In the AI Interview Platform (PrepWise), testers interact with the platform by assuming the roles of students, fresh graduates, and recruiters to simulate real-world interview preparation scenarios. This testing approach allows evaluators to closely observe system behavior from an end-user perspective.

Manual testing is used to validate key functionalities such as authentication (login/signup), dashboard navigation, AI interview generation, real-time mock interviews, feedback reporting, multiple retakes, and recruiter search workflows. Since this method focuses on actual user interaction, it is particularly effective in identifying usability issues, interface inconsistencies, unexpected errors, and broken user flows. However, while manual testing excels at detecting surface-level and experience-related problems, certain internal logic or performance-related issues may require deeper inspection through automated or technical testing methods.

6.1.1 System Testing

System testing is performed to verify that the complete AI Interview Platform functions correctly as an integrated and unified system. This phase ensures that individual modules such as user authentication, dashboard management, AI interview generation, real-time mock interviews, feedback reporting, multiple retakes, recruiter search and ranking, and adaptive learning features work together seamlessly without conflicts or data inconsistencies.

After the completion of development cycles, system testing evaluates whether the overall platform meets specified functional requirements, performance benchmarks, and quality standards. It focuses on end-to-end workflows to confirm that data flows smoothly across components and that user actions produce expected outcomes. Successful system testing provides confidence that the platform is stable, reliable, and ready for deployment, ensuring a consistent and high-quality experience for all users .

6.1.2 Unit Testing

Unit testing in the AI Interview Platform (PrepWise) focuses on verifying each functional component independently to detect issues at the smallest level. Core units tested include user login and authentication, dashboard navigation, AI interview generation, feedback reporting, recruiter search filters, and multiple retake functionality. For example, tests ensure that user input (such as email, password, or interview parameters) meets expected formats, that dashboard sections save and update correctly, and that AI modules return valid outputs when provided with candidate skills or interview responses.

Unit tests are repeatable and executed frequently during development to ensure that updates or code changes do not break existing functionality. This module-level testing enables developers to maintain a stable foundation before proceeding to integration, system, and acceptance testing.

Unit Testing 1: Authentication and User Management

Testing Objective: To ensure that the login, signup, password reset, and third-party authentication functionalities work correctly and securely as shown in table 6.1.

Table 6.1 UT 1 Authentication and User Management

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
AUTH-01	Login with valid credentials	Email + Correct Password	Login success, dashboard displayed	Login success, dashboard displayed	Pass
AUTH-	Login with	Email +	“Invalid	“Invalid	Pass

02	invalid password	Wrong Password	credentials" error	credentials" error	
AUTH-03	Signup with missing fields	Email only	Field validation errors	Field validation errors	Pass
AUTH-04	Google OAuth login test	Google token	Successful login	Successful login	Pass
AUTH-05	Password reset request	Registered email	Reset link sent	Reset link sent	Pass

Unit Testing 2: Interview Generation Module

Testing Objective: o ensure that interview sessions can be created, customized, started, and stored correctly in the system as shown in table 6.2.

Table 6.2 UT 2 Interview Generation Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
INT-01	Creating a new interview session	Valid role + level + tech stack	Interview generated successfully	Interview generated successfully	Pass
INT-02	Edit interview parameters	Updated role/stack details	Interview updated	Interview updated	Pass
INT-03	Delete an interview session	Interview ID	Interview removed	Interview removed	Pass
INT-04	Apply predefined template	Template ID	Template applied correctly	Template applied correctly	Pass
INT-05	Save candidate responses	Candidate answers	Responses stored accurately	Responses stored accurately	Pass

Unit Testing 3: Dashboard Module

Testing Objective: To ensure dashboard displays interviews and allows access to shared sessions.as given in table 6.3.

Table 6.3 UT 3 Dashboard Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
DASH-01	View own interviews	User ID	List of interviews displayed	List of interviews displayed	Pass
DASH-02	Access shared interview	Shared Interview ID	Interview opened successfully	Interview opened successfully	Pass

DASH-03	Invalid interview ID	Wrong ID	Error message	Error message	Pass

Unit Testing 4: Real-time Mock Interview

Testing Objective: To ensure AI interviewer asks questions and records candidate answers. in table 6.4.

Table 6.4 UT 4 Recruiter Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
MOCK-01	Start mock interview	Candidate ID	Interview session started	Interview session started	Pass
MOCK-02	Record candidate response	Candidate answer	Response saved	Response saved	Pass
MOCK-03	Invalid session ID	Wrong ID	Error message	Error message	Pass

Unit Testing 5: Feedback System

Testing Objective: To ensure that feedback reports are generated accurately based on user interview responses in table 6.5.

Table 6.5 UT 5 Feedback System

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
FB-01	Generate feedback report	Candidate interview session	Report generated successfully	Report generated successfully	Pass
FB-02	Missing interview data	Null/empty responses	Error message returned	Error message returned	Pass
FB-03	Communication analysis	Candidate answers	Communication score generated	Communication score generated	Pass
FB-04	Confidence evaluation	Candidate answers	Confidence score generated	Confidence score generated	Pass
FB-05	Large dataset performance	1000 responses	Report generated without delay	Report generated without delay	Pass

Unit Testing 6: Multiple Retakes

Testing Objective: To ensure candidates can retake interviews and improve scores. as given in table 6.6.

Table 6.6 UT 6 Multiple Retakes

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
RET-01	Retake interview	Interview ID	New attempt created	New attempt created	Pass
RET-02	Compare scores	Candidate ID	Updated score displayed	Updated score displayed	Pass

6.1.3 Functional Testing

Functional testing in the AI Interview Platform (PrepWise) is carried out after unit testing to ensure that every module behaves exactly as defined in the system requirements. The major user roles Recruiter and Candidate are tested to confirm that their respective features load correctly, display accurate data, and respond as expected to user actions.

Functional Testing 1: Login (Recruiter & Candidate Roles)

Objective: To ensure that both **Recruiter** and **Candidate** can log in securely, are redirected to their respective dashboards, and proper validation and logout functionality works, given in table 6.7.

Table 6.7 FT 1 Login (Recruiter & Candidate Roles)

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
FT-001	Login with valid candidate credentials	Email: candidate@example.com Password: ****	Candidate dashboard loads with interview history and options	Candidate dashboard loads with interview history and options	Pass
FT-002	Login with valid recruiter credentials	Email: recruiter@example.com Password: ****	Recruiter dashboard opens showing job postings and applications	Recruiter dashboard opens showing job postings and applications	Pass
FT-003	Login with invalid credentials	Wrong email or wrong password	System displays “Invalid credentials”	System displays “Invalid credentials”	Pass
FT-004	Login with empty fields	Email: empty Password: empty	Validation message displayed: “All fields	Validation message displayed: “All fields	Pass

			are required"	are required"	
FT-005	Logout	User clicks logout button	User returns to login screen and session ends	User returns to login screen and session ends	Pass

The functional testing table for the login process evaluates the behavior of the PrepWise authentication system across both user categories Recruiters and Candidates. Test Case FT-001 verifies the candidate login flow by checking whether valid credentials correctly redirect the candidate to a personalized dashboard displaying their interview history and available options. Following this, Test Case FT-002 ensures that recruiters logging in with valid details are accurately directed to their designated dashboard, which includes job postings and application tracking. Test Case FT-003 examines the system's ability to detect invalid login attempts by providing an incorrect email or password; the expected "Invalid credentials" message is displayed, indicating proper error handling. Test Case FT-004 ensures robust validation by testing empty fields, where the system appropriately prompts the user that all fields are required. Finally, Test Case FT-005 confirms that the logout feature terminates the user session securely and redirects them back to the login interface. All test cases passed successfully, demonstrating that AI Interview Platform authentication module consistently delivers secure access control and accurate dashboard routing for both user types**.**

Functional Testing 2: Candidate Module

Objective: To verify that candidates can manage their interview profiles, generate and attempt AI-based mock interviews, receive feedback, retake session as given in table 6.8.

Table 6.8 FT 2 Candidate Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
FT-C001	View Dashboard	Candidate logs in and navigates to dashboard	Dashboard shows interview history, shared sessions, and retake options	Dashboard shows interview history, shared sessions, and retake options	Pass
FT-C002	Generate AI interview	Inputs: Role, Level, Type, Tech stack	New interview session created and displayed in dashboard	New interview session created and displayed in dashboard	Pass
FT-C003	Update candidate profile	Name, picture, skill details	Profile updated and reflected instantly	Profile updated and reflected instantly	Pass
FT-C004	Attempt real-time mock	Candidate answers	AI interviewer records responses	AI interviewer records responses	Pass

	interview		and session saved	and session saved	
FT-C005	Receive feedback report	Interview session completed	Feedback generated (communication, confidence, problem-solving)	Feedback generated (communication, confidence, problem-solving)	Pass
FT-C006	Retake interview	Interview ID	New attempt created and updated score displayed	New attempt created and updated score displayed	Pass

The functional testing table for the Candidate Module evaluates the essential features that support smooth interview preparation and career-building experience within AI Interview Platform. Test Case FT-C001 verifies that the candidate dashboard loads properly after login, displaying personalized interview history, shared sessions, and retake options. Test Case FT-C002 checks the AI Interview Generator functionality, confirming that inputs such as role, level, type, and tech stack produce accurate interview sessions. Test Case FT-C003 validates the profile editing mechanism, ensuring updates to the candidate's personal details and skills are instantly reflected. In Test Case FT-C004, the real-time mock interview process is assessed, and the system correctly records candidate responses and saves the session. Test Case FT-C005 evaluates the feedback system by ensuring that communication, confidence, and problem-solving scores are generated based on candidate performance. Test Case FT-C006 confirms that candidates can retake interviews, with the system creating new attempts and updating scores accordingly. All tests passed successfully, indicating that the Candidate Module functions reliably, supports seamless interview preparation, and provides meaningful feedback and career-related interactions.

Functional Testing 3: AI Interview Generator Module

Objective: To ensure that candidates can generate customized interview sessions, edit parameters, delete sessions, switch templates, and apply AI formatting consistently across all interview types as shown in table 6.9.

Table 6.9 FT 3 AI Interview Generator Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
FT-I001	Add interview parameters	Role, Level, Tech stack	Interview generated with selected parameters	Interview generated with selected parameters	Pass
FT-I002	Edit interview	Updated Role/Level/Tech	Interview session updated	Interview session updated	Pass

	session	stack	and displayed in preview	and displayed in preview	
FT-I003	Delete interview session	Selected session removed	Session deleted with confirmation	Session deleted with confirmation	Pass
FT-I004	Template switching	Switch between Template A/B/C	Interview preview updates instantly	Interview preview updates instantly	Pass
FT-I005	Auto-format interview	One-click AI formatting	Consistent layout applied across all interview questions	Consistent layout applied across all interview questions	Pass

The functional testing table for the AI Interview Generator Module ensures that candidates can construct, customize, and manage their interview sessions with accuracy and consistency. Test Case FT-I001 validates the addition of interview parameters by allowing candidates to enter role, level, and tech stack, after which the system successfully generates a tailored interview. Test Case FT-I002 checks the editing functionality; when candidates update parameters such as role or level, the system correctly refreshes the interview session preview. Test Case FT-I003 examines the deletion function, where a selected interview session is removed only after user confirmation, ensuring accidental deletions are prevented. In Test Case FT-I004, the system is tested for responsiveness when switching between multiple interview templates (A, B, or C), and results show that the preview updates instantly without layout issues. Finally, Test Case FT-I005 evaluates the auto-formatting tool, which applies a unified and consistent design across all interview questions in a single action, ensuring professional presentation. All tests achieved a “Pass,” demonstrating that AI Interview Generator operates reliably and meets candidate expectations for customized interview preparation

.**Functional Testing 4:** Real-time Mock Interview Module

Objective: To ensure that candidates can attempt AI-driven mock interviews, where the system asks questions, records responses, and saves sessions reliably,as shown in table 6.10.

Table 6.10 FT 4 Real-time Mock Interview Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
FT-M001	Start mock interview	Candidate ID	Interview session starts with AI interviewer	Interview session starts with AI interviewer	Pass
FT-M002	AI asks questions	Predefined interview	AI interviewer displays	AI interviewer displays	Pass

		template	questions sequentially	questions sequentially	
FT-M003	Candidate provides response	Candidate answers	System records and saves responses	System records and saves responses	Pass
FT-M004	End interview session	Candidate clicks “End”	Session ends and saved in dashboard	Session ends and saved in dashboard	Pass
FT-M005	Invalid session ID	Wrong ID	Error message displayed	Error message displayed	Pass

The functional testing table for the Real-time Mock Interview Module evaluates the behavior of the AI Interview Platform platform during live interview simulations. Test Case FT-M001 verifies that candidates can successfully start a mock interview session with the AI interviewer. Test Case FT-M002 ensures that the AI interviewer displays questions sequentially according to the selected template. Test Case FT-M003 validates that candidate responses are recorded and saved accurately. In Test Case FT-M004, the system is tested for proper termination of the interview session, confirming that the session ends securely and is stored in the candidate’s dashboard. Finally, Test Case FT-M005 examines error handling when an invalid session ID is provided, ensuring that the system displays the correct error message. All test cases passed successfully, demonstrating that the Real-time Mock Interview Module delivers reliable, interactive interview simulations and supports seamless candidate participation**.**

Functional Testing 5: Feedback System Module

Objective: To ensure that candidates receive accurate feedback reports after completing interviews, covering communication, confidence, problem-solving, and improvement areas as shown in table 6.11.

Table 6.11 FT 5 Feedback System Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
FT-F001	Generate feedback report	Completed interview session	Feedback report generated with scores and suggestions	Feedback report generated with scores and suggestions	Pass
FT-F002	Missing interview data	Null/empty responses	Error message displayed	Error message displayed	Pass
FT-F003	Confidence evaluation	Candidate answers	Confidence score calculated and displayed	Confidence score calculated and displayed	Pass
FT-F004	Communication analysis	Candidate responses	Communication score generated	Communication score generated	Pass

FT-F005	Problem-solving evaluation	Candidate responses	Problem-solving score generated	Problem-solving score generated	Pass
FT-F006	Improvement suggestions	Candidate performance data	System provides targeted improvement recommendations	System provides targeted improvement recommendations	Pass

The functional testing table for the Feedback System Module evaluates the behavior of the AI Interview platform in generating performance insights for candidates. Test Case FT-F001 verifies that a feedback report is generated after the completion of an interview, including scores and improvement suggestions. Test Case FT-F002 ensures that the system handles missing or empty interview data by displaying the appropriate error message. Test Case FT-F003 validates the confidence evaluation mechanism, confirming that candidate responses are analyzed to produce a confidence score. Test Case FT-F004 checks the communication analysis feature, ensuring that the system generates accurate communication scores. Test Case FT-F005 examines the problem-solving evaluation, confirming that candidate answers are assessed for analytical and logical skills. Finally, Test Case FT-F006 validates that the system provides targeted improvement suggestions based on candidate performance. All test cases passed successfully, demonstrating that the Feedback System Module delivers reliable, actionable insights and supports continuous candidate improvement.

6.1.4 Integration Testing

Integration testing in the AI Interview Platform (PrepWise) is a critical phase that focuses on verifying the interactions between individual modules, such as the Authentication system, Candidate Dashboard, AI Interview Generator, Real-time Mock Interview, Feedback System, and Multiple Retakes [25]. This testing ensures that these components communicate correctly, exchange data seamlessly, and work together as intended.

By detecting interface inconsistencies, logical errors, and potential data mismatches, integration testing validates that the platform functions smoothly in real-world scenarios, providing reliable authentication, accurate AI-driven interview generation, consistent feedback analysis, and seamless retake workflows. This phase enhances the overall system reliability and prepares PrepWise for subsequent testing and deployment stages.

Integration Testing 1: Login as Candidate

Objective: To ensure that all candidate-related modules including authentication, dashboard, interview generation, mock interview, feedback, and retake handling interact seamlessly, enabling correct access, updates, and secure logout as shown in table 6.12.

Table 6.12 IT 1 Login as Admin

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
IT-001	Candidate login + dashboard access	Authentication Module + Dashboard Module	Candidate logs in and the dashboard loads with interview history	Candidate logs in and the dashboard loads with interview history	Pass
IT-002	Generate interview + dashboard update	AI Interview Generator + Dashboard Module	New interview generated and displayed in dashboard	New interview generated and displayed in dashboard	Pass
IT-003	Attempt mock interview + feedback	Mock Interview Module + Feedback Module	Candidate attempts interview, feedback generated automatically	Candidate attempts interview, feedback generated automatically	Pass
IT-004	Retake interview + updated scores	Multiple Retakes Module + Feedback Module	New attempt created, updated scores shown in the dashboard	New attempt created, updated scores shown in the dashboard	Pass
IT-005	Candidate logout	Dashboard Module + Session Module	Candidate logged out, and the session terminated	Candidate logged out, and the session terminated	Pass

The integration testing for the Candidate workflow in PrepWise ensures that critical modules such as authentication, dashboard, interview generation, mock interview, feedback, and retakes work smoothly together. Test Case IT-001 confirms that after the candidate logs in through the authentication module, the system correctly hands off privileges to the dashboard module, enabling successful access to interview history. Test Case IT-002 validates the integration between the AI Interview Generator and dashboard; when a candidate generates a new interview, the system not only creates the session but also updates the dashboard instantly. In Test Case IT-003, the integration between the mock interview and feedback modules is tested, confirming that responses recorded during the interview are analyzed and feedback is generated automatically. Test Case IT-004 evaluates the retake functionality, ensuring that new attempts are created and updated scores are reflected in the dashboard and feedback system. Finally, Test Case IT-005 checks proper integration between the dashboard and session management, ensuring smooth logout and secure session termination.

All integration points passed successfully, demonstrating stable connectivity across modules. Additionally, these tests confirm that the candidate interface responds efficiently to concurrent operations, maintaining data integrity even under multiple simultaneous actions. The seamless coordination between modules highlights the robustness of the AI interview platform system architecture, ensuring reliable performance for candidate workflows across the platform.

Integration Testing 2: AI Interview Generator Module

Objective: To ensure that all candidate-related modules including authentication, dashboard, interview generation, mock interview, feedback, and retake handling interact seamlessly, enabling correct access, updates, and secure logout in Table 6.13.

Table 6.13 IT 2 AI Interview Generator Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
IT-101	Candidate login + dashboard access	Authentication Module + Dashboard Module	Candidate logs in, and the dashboard loads with interview history	Candidate logs in, and the dashboard loads with interview history	Pass
IT-102	Generate interview + dashboard update	AI Interview Generator + Dashboard Module	New interview generated and displayed in the dashboard	New interview generated and displayed in the dashboard	Pass
IT-103	Attempt mock interview + feedback	Mock Interview Module + Feedback Module	Candidate attempts interview, feedback generated automatically	Candidate attempts interview, feedback generated automatically	Pass
IT-104	Retake interview + updated scores	Multiple Retakes Module + Feedback Module	New attempt created, updated scores shown in the dashboard	New attempt created, updated scores shown in the dashboard	Pass
IT-105	Candidate logout	Dashboard Module + Session Module	Candidate logged out, and the session terminated	Candidate logged out, and the session terminated	Pass

Integration Testing 3: Recruiter Module

Objective: To ensure that all recruiter-related modules including authentication, dashboard, job posting, candidate application management, and session handling interact seamlessly, enabling correct access, updates, and secure logout as given in Table 6.14.

Table 6.14 IT 3 Recruiter Module

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
IT-201	Recruiter login + dashboard access	Authentication Module + Recruiter Dashboard	The recruiter logs in, and the dashboard loads with job analytics	The recruiter logs in, and the dashboard loads with job analytics	Pass
IT-202	Create new job posting	Recruiter Dashboard + Job Posting Module	New job posting created and visible to candidates	New job posting created and visible to candidates	Pass
IT-203	View candidate applications	Recruiter Dashboard + Applications Module	Candidate applications are displayed with details	Candidate applications are displayed with details	Pass
IT-204	Recruiter logout	Recruiter Dashboard + Session Module	The recruiter logged out, and the session terminated	The recruiter logged out, and the session terminated	Pass

The integration testing for the Recruiter workflow in PrepWise ensures that critical modules such as authentication, the recruiter dashboard, and candidate applications work smoothly together. Test Case IT-201 confirms that after the recruiter logs in through the authentication module, the system correctly hands off privileges to the recruiter dashboard, enabling successful access to job analytics. Test Case IT-202 validates the integration between the recruiter dashboard and job posting module; when a recruiter creates a new job, the system not only saves the posting but also makes it visible to candidates instantly. In Test Case IT-203, the integration between the recruiter dashboard and applications module is tested, confirming that candidate applications are displayed with complete details. Finally, Test Case IT-205 checks proper integration between the recruiter dashboard and session management, ensuring smooth logout and secure session termination.

All integration points passed successfully, demonstrating stable connectivity across recruiter modules. Additionally, these tests confirm that the recruiter interface responds efficiently to concurrent operations, maintaining data integrity even under multiple simultaneous actions. The seamless coordination between modules highlights the robustness of the AI interview system architecture, ensuring reliable performance for recruiter workflows across the platform.

Integration Testing 4: AI Interview Generator Workflow

Objective: To ensure that the AI Interview Generator integrates seamlessly with authentication, dashboard, mock interview, feedback, and retake modules, enabling candidates to generate, attempt, and review interviews correctly as shown in Table 6.15.

Table 6.15 IT 4 AI Interview Generator Workflow

Test Case ID	Test Description	Input	Expected Output	Actual Output	Remarks
IT-301	Generate interview + dashboard update	Authentication Module + AI Interview Generator + Dashboard	The candidate generates an interview, and the dashboard updates instantly	The candidate generates an interview, and the dashboard updates instantly	Pass
IT-302	Edit interview parameters	AI Interview Generator + Dashboard Module	Updated interview session reflected in dashboard	Updated interview session reflected in dashboard	Pass
IT-303	Attempt generated interview	AI Interview Generator + Mock Interview Module	Candidate attempts generated and interview, and responses were recorded	Candidate attempts generated and interview, and responses were recorded	Pass
IT-304	Feedback after the generated interview	AI Interview Generator + Feedback Module	Feedback was generated based on the candidate’s responses	Feedback was generated based on the candidate’s responses	Pass
IT-305	Retake the generated interview	AI Interview Generator + Multiple Retakes Module	New attempt created, updated scores shown in the dashboard	New attempt created, updated scores shown in the dashboard	Pass

The integration testing for the AI Interview Generator workflow in PrepWise ensures that modules such as authentication, dashboard, mock interview, feedback, and retakes work smoothly together. Test Case IT-301 confirms that after the candidate logs in and generates an interview, the system correctly updates the dashboard with the new session. Test Case IT-302 validates the integration between the generator and dashboard; when interview parameters are edited, the updated session is reflected instantly. In Test Case IT-303, the integration between the generator and mock interview module is tested, confirming that generated interviews can be attempted and responses recorded accurately. Test Case IT-304 evaluates the feedback mechanism, ensuring that performance insights are generated based on candidate responses in generated interviews. Finally, Test Case IT-305 checks proper integration with the retake module, confirming that candidates can retake generated interviews and view updated scores in the dashboard.

All integration points passed successfully, demonstrating stable connectivity across modules. These tests confirm that the AI Interview Generator Module supports seamless candidate workflows, and ensures reliable preparation for interviews.

6.2 Tools and Technologies

Table 6.16 The development of the **PrepWise – AI Interview Platform** is supported by a modern technology stack that ensures scalability, security, and real-time performance. On the frontend, **Next.js** and **React.js** are used to build dynamic and responsive user interfaces, while **Tailwind CSS** provides customizable styling and **TypeScript** enhances code reliability through type safety. For backend operations, **Firebase Authentication** manages secure login and role-based access, and **Firestore Database** stores candidate profiles, interview sessions, and feedback reports with real-time synchronization. The platform integrates **Voice AI protocols (VAPI)** and **Natural Language Processing (NLP)** techniques to simulate interactive interviews and analyze candidate responses. Machine learning models are employed to generate predictive scoring and feedback on communication, confidence, and problem-solving skills. Testing is conducted through frameworks such as **React Testing Library** for unit testing, alongside integration and manual usability testing to validate module interactions. Deployment is handled via **Firebase Hosting**, ensuring secure and scalable access, Collectively, these tools and technologies enable PrepWise to deliver a robust, intelligent, and user-friendly platform for AI-driven interview preparation.

Table 6.16 Tools and Technologies

Tools & Technologies	Version	Rationale
Next.js	Latest	Server-side rendering and optimized frontend performance
React.js	Latest	Dynamic UI components and smooth candidate/recruiter interactions
Tailwind CSS	Latest	Responsive and customizable styling across devices
TypeScript	Latest	Type safety and maintainability of frontend code
Firebase Authentication	Latest	Secure login and role-based access for candidates and recruiters
Firebase Firestore	Latest	Real-time database for storing interview sessions and feedback reports

VAPI (Voice AI Protocols)	Latest	Voice-based interview simulations and backend integration
REST APIs	Latest	Smooth communication between frontend and backend modules
NLP (Natural Language Processing)	Latest	Analyze candidate responses during mock interviews
Machine Learning Models	Latest	Predictive scoring and feedback generation (confidence, communication, problem-solving)
React Testing Library	Latest	Unit testing of components and modules
Integration Testing Tools	Latest	Validate module interactions (authentication, dashboard, feedback, retakes)
Manual Usability Testing	Latest	Ensure smooth navigation and user experience
Firebase Hosting	Latest	Secure and scalable hosting for the platform

6.3 Summary

This chapter presents the various testing approaches applied to the PrepWise – AI Interview Platform to ensure system reliability, correctness, and overall quality. The testing strategy includes unit testing, functional testing, and integration testing, each targeting different levels of system validation. All major modules such as authentication and login, candidate dashboard, AI interview generator, real-time mock interview, feedback system, and multiple retakes were systematically evaluated using well-structured test cases.

These test cases were designed to cover both standard operational scenarios and critical edge-case conditions to verify robustness under diverse candidate behaviors and data inputs. Through this comprehensive testing process, the platform's functional accuracy, module interoperability, and user experience were thoroughly assessed, ensuring that the system meets defined requirements and performs consistently across all use cases.

Chapter 7
Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The AI Interview Platform is designed to streamline the preparation, practice, and evaluation of interviews by offering intelligent insights to candidates and recruiters. The primary objective of this project was to develop a comprehensive digital ecosystem where users can generate customized interview sessions, attempt real-time mock interviews, and receive AI-powered feedback, all within a single platform. For recruiters, PrepWise provides efficient tools to review candidate applications, and assess interview performance, while candidates gain personalized guidance and actionable suggestions for improvement.

PrepWise has successfully transformed the interview preparation process, making it more organized, interactive, and user-friendly. Traditionally, candidates relied on static question banks, manual practice, and scattered feedback, which were often inefficient and lacked personalization. With PrepWise, all relevant interview data, performance history, and feedback are centralized, digitally accessible, and continuously updated through the dashboard.

A significant advantage of PrepWise is its intuitive and responsive interface. Users do not need technical expertise to navigate the platform. Whether generating interviews, attempting mock sessions, or reviewing feedback, candidates and recruiters can interact with the system effortlessly. The platform ensures seamless navigation, clear workflows, and a smooth user experience across all modules.

During development and testing, rigorous quality assurance was performed on features including authentication, interview generation, mock interview handling, feedback analysis, and multiple retakes. All modules have been thoroughly tested and operated as intended, ensuring reliability and consistency.

In conclusion, PrepWise effectively achieves its purpose of enhancing interview preparation while delivering intelligent, data-driven recommendations. It consolidates scattered processes, simplifies interview practice, and increases visibility for candidate performance. By combining user-friendly design, AI-powered insights, and recruiter tools, PrepWise offers a holistic solution that is both practical and forward-looking. It represents a significant advancement in integrating technology into professional development, making interview preparation more accessible, actionable, and impactful for all stakeholders. The platform addresses existing challenges in interview readiness while laying the groundwork for future enhancements, including advanced analytics, personalized learning pathways, and improved recruiter-candidate matching, further cementing its role in the professional landscape.

7.2 Future Work

Although the current version of the AI Interview Platform (PrepWise) is fully functional and successfully meets its objectives, there remains substantial scope to enhance its features, usability, accessibility, and overall user experience. The future development of PrepWise aims to make the platform smarter, more interactive, and inclusive, while ensuring strong data security and seamless performance.:

7.2.1 Mobile Applications (Android and iOS)

Developing dedicated mobile applications for both Android and iOS platforms will allow candidates and recruiters to access the AI Interview platform anytime and anywhere. The mobile apps will provide a fully responsive interface for interview generation, mock practice, AI-driven feedback, real-time notifications for job postings and application updates, and quick communication between candidates and recruiters. Offline features, such as cached interview sessions and draft feedback storage, can also be integrated to enhance accessibility in areas with limited internet connectivity.

7.2.2 Advanced AI Features

Future enhancements will focus on expanding the AI engine to deliver deeper and more actionable insights. This includes personalized skill gap analysis to identify areas for improvement, automated candidate ranking based on multiple criteria such as communication, confidence, and technical skills, and career path recommendations tailored to individual profiles.

7.2.3 Enhanced Interview Customization

To improve engagement and personalization, candidates will be given more options to fully customize their interview sessions. Features may include custom templates, adjustable difficulty levels, interactive question formats, and dynamic visualization of performance analytics. This will allow candidates to tailor interview practice to their career goals, enabling greater flexibility and professional branding.

7.2.4 Camera Integration

AI Interview Platform**,** camera integration can enable direct face-to-face interviews between candidates and recruiters. This feature will allow real-time interaction within the platform, eliminating reliance on external tools. The AI engine will analyze both verbal responses and non-verbal cues such as facial expressions and gestures. Strong security protocols will ensure encrypted video sessions and protect sensitive data. Overall, camera-enabled interviews will make PrepWise a hybrid solution for AI-driven practice and real-world recruitment.[17]

7.2.5 Strong Security and Privacy Measures

Future work will emphasize advanced security and privacy protocols to protect sensitive candidate and recruiter data. This includes two-factor authentication (2FA), end-to-end encryption for interview recordings and feedback reports, role-based access control, and compliance with data privacy regulations such as GDPR. These measures will ensure that interview sessions, personal information, and communications remain secure from unauthorized access, maintaining user trust and platform integrity.

7.2.6 Performance and Scalability Optimization

As the platform grows in user base and data volume, performance optimization will be essential. Future improvements will focus on enhancing system speed, reducing loading times, and ensuring smooth responsiveness during intensive operations such as AI computations and real-time mock interviews. Scalability enhancements will allow the platform to handle a growing number of concurrent users, extensive interview datasets, and real-time analytics without lag or downtime, ensuring a seamless experience for all stakeholders.

7.2.7 Integration with External Academic and Professional Platforms

Future iterations may include integration with platforms such as LinkedIn, GitHub, and online job portals to automatically import candidate profiles, certifications, and professional experiences. This will further reduce manual data entry, keep profiles up to date, and enrich AI recommendations for both candidates and recruiters.

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